

Prices vs. Production Restrictions: China's Coal Industry*

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Abstract

Governments often use command-and-control production restrictions to reduce pollution. Unlike taxes, these restrictions need not reduce output according to producer efficiency and can therefore misallocate production. We show theoretically that, with shocks to the economic environment, the welfare ranking of a pollution tax and a production restriction is ambiguous. We then estimate an interprovincial model of coal demand and supply in China and use it to study the 2016 industry reform. Counterfactual simulations show that the observed policy is less efficient than a tax, but can be rationalized by a policy preference for producer surplus.

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1 Introduction

Regulators often use price- and quantity-based policies to mitigate externalities. Economists have long advocated for market-based instruments, such as Pigouvian taxes (Pigou, 1920; Baumol, 1972) or cap-and-trade systems (Crocker, 1966; Dales, 1968; Montgomery, 1972), to internalize externalities efficiently. Although resource taxation is common, sophisticated cap-and-trade markets are often infeasible due to high transaction costs (Nie, 2012) or weak institutional capacity (Bell and Russell, 2002; Blackman and Harrington, 2000), especially in developing countries. In these cases, regulators may resort to directly restricting production, such as factory shutdowns or moratoria on resource extraction (Blackman, Li and Liu, 2018).

We study a major reform of China’s coal industry launched in 2016 that restricted coal production. Our goal is to compare the welfare effects of the restriction with those of a tax. Policy choices in this setting are consequential. As the world’s largest producer and consumer of coal, China faces acute tensions between energy security, environmental consequences, and economic stability. It has been widely reported that the reliance on coal undermines China’s energy transition and global efforts to mitigate climate change (McGrath, 2019; WSJ, 2022; Bloomberg, 2022). A number of papers have documented the severe air-quality and health consequences associated with coal (Almond et al., 2009; Chen et al., 2013; Ebenstein et al., 2017). At the same time, coal mining provides significant employment opportunities and local government revenues, especially in major producing provinces dominated by state-owned enterprises (Wright, 2012; Hervé-Mignucci et al., 2015). Policies affecting coal output therefore have important distributional consequences (Clark and Zhang, 2022; Yuan et al., 2025).

As part of a multi-year effort to reduce coal consumption and pollution while preserving the industry’s employment and profitability, the Chinese government in April 2016 adopted a production restriction policy, ordering small mines to close and large mines to reduce output. As provinces implemented these measures, coal prices surged: our data show that the average coal price in the second half of 2016 rose by 31.0% to RMB 539 (\$77.4)/ton, from RMB 411 (\$60)/ton in the same period of 2015. The government continued to close mines and strictly monitor production to curtail output above approved capacity at least through 2018.

From a welfare perspective, why might the government prefer this policy to taxation? We analyze this question in a setting with uncertainty in demand, supply, and pollution damages. Using a local approximation based on Weitzman (1974), we first show that it is theoretically ambiguous whether a pollution tax or a production restriction achieves higher social welfare. Second, we show how the preferred policy may change when the government places a greater weight on producer surplus and pollution damages. Third, we propose a

criterion for comparing the two policies when researchers observe either the benefits or the costs of abatement, but not both. The key idea is based on stochastic dominance. Suppose, for example, that we can measure abatement costs under both policies and that higher quantities generate weakly higher pollution damages. If one policy has higher abatement costs (more lost economic surplus) and also induces a stochastically dominating distribution of quantity (more pollution), then the policy must imply lower social welfare.

To empirically analyze the demand and supply in the coal market, we combine the policy intervention with a comprehensive dataset on coal production and consumption. Our main data set includes monthly prices at the province level and shipment quantities from producers to end users both within and across provinces. We validate the shipment data against production and consumption data from other sources.

We next formulate a model that incorporates key institutional features of China’s massive and partially regulated coal market. First, coal procurement in the electricity and other sectors has undergone significant market reforms since 2013, essentially allowing the market to set prices that equilibrate demand and supply. Second, the shares of coal shipped from one province to another are highly stable over time, despite varying price dispersions across provinces. In our model, the market is competitive. We specify a demand function and a supply function for each province. The local price in each province is determined by the province’s demand and the total supply from different provinces. The quantity supplied by a province is determined by a weighted average of prices in provinces it sells to. The effects of the policy are modeled as a shift in the supply curve that depends on production capacity.

We estimate the demand and supply of coal across provinces. To identify demand, we construct price instruments using variations in policy compliance levels across provinces that shift supply. The key identifying assumption is that the policy exclusively affects supply and is not correlated with demand unobservables. We estimate a demand elasticity of about -0.6 , which is robust across a variety of alternative instruments. Similarly, we estimate supply using demand shifters. The estimates from our preferred specification suggest that the policy significantly steepens aggregate marginal costs once production approaches capacity.

Given the estimated equilibrium model of coal demand and supply, we conduct a series of counterfactual simulations to quantify the welfare effects of restricting production. The baseline welfare measure accounts for changes in economic surplus, tax revenue, and pollution damages, which include damages from air pollution and the environmental costs of coal production.

First, we find that removing the restrictions would increase annual production by 668 million tons and decrease the price by RMB 145/ton. Economic surplus would increase

by RMB 259 billion, mostly through consumer surplus gains from lower coal prices, but pollution damages would also increase by RMB 309 billion. Social welfare is therefore RMB 50 billion higher under the observed production restriction than under no policy.

Second, we compare the policy with an efficient quota or a cap-and-trade system to understand its distortionary effects. Our supply estimates suggest that the policy increases aggregate marginal costs, especially near approved capacity, generating inefficient reductions in quantity both within and across provinces. Holding fixed each province's production under the observed policy, removing the within-province distortion would raise producer surplus by RMB 136 billion. Efficiently reallocating quantity reductions across provinces would further increase the producer surplus gain to RMB 212 billion.

We then use the estimates to rank the production restriction policy against a tax. We find that an optimal tax of RMB 417/ton would raise baseline welfare by RMB 261 billion relative to the observed policy. Imposing an optimal ad valorem tax has similar effects. Next, we generalize the baseline welfare function to include an unknown pollution damage function that increases in total quantity. This damage can reflect the social cost of carbon, but more generally captures any social costs that we do not quantify but that rise with aggregate production. Using the criterion proposed in the theory section, we show that the tax above can achieve higher welfare for any such damage function.

Finally, motivated by the stated policy goals of improving industry profits and protecting the environment, we ask whether the observed policy can be rationalized by a policy objective function that places different weights on producer surplus and avoided pollution damages. Specifically, we construct a weighted welfare function and compare its value under the observed production restriction with its value under the optimal tax chosen with the same weights. We find that, when the weight on avoided pollution damages is one, the observed policy would outperform even the optimal pollution tax if the government values producer surplus 40.8% more than other components of welfare. The intuition is that the weighted loss of producer surplus under a tax can be so large that tax revenue cannot offset it, whereas production restrictions can directly increase producer surplus. We also find that a greater weight on avoided pollution damages must be paired with an even greater preference for producer surplus for the government to prefer the observed policy to a tax, because taxes tend to reduce quantities more for the same producer surplus.

Related Literature and Contribution

We first contribute to a literature that uses observed policy choices to understand political or social preferences. In trade, Goldberg and Maggi (1999) estimate the government’s relative weight on aggregate welfare versus political contributions. A number of studies examine political-economy models of trade protection (Gawande and Bandyopadhyay, 2000; Mitra, Thomakos and Ulubaşoğlu, 2002; McCalman, 2004; Matschke and Sherlund, 2006; Gawande, Krishna and Olarreaga, 2009; Francois and Nelson, 2014; Adão et al., 2026). Other related studies estimate social welfare weights that rationalize observed tax-benefit systems (Bourguignon and Spadaro, 2012; Bargain et al., 2014; Hendren, 2020) and infer political motives from infrastructure choices, including rail in California and highways in China (Fajgelbaum et al., 2024; Alder and Kondo, 2019). Relative to this literature, which typically infers preferences from the intensity of a given policy instrument (such as tariff rates or tax levels), we use the government’s choice between instruments, the observed production restriction vs. tax, to bound the relative weights on components of the welfare function.

We build on the classic framework of Weitzman (1974) to compare taxation with production restriction.¹ Weitzman (1974) studies the choice between price and quantity instruments under uncertainty, while a set of subsequent work compares taxes with permits, standards, command-and-control rules, and hybrid policies (Adar and Griffin, 1976; Roberts and Spence, 1976; Koenig, 1985; Helfand, 1991; Dickie and Trandel, 1996). Our comparison differs in two respects. First, we relate the welfare comparison to policy preferences and show how they interact. Second, our counterfactual analysis simulates the market equilibrium under alternative policies, going beyond a theoretical local approximation. We also develop a criterion to compare the equilibrium welfare effects of instruments, even when the benefits of abatement are not fully measured. Our approach uses estimated demand and supply models to compute market equilibrium and welfare, which differs from prior work comparing policy instruments that uses integrated assessment models (Pizer, 2002; Muller and Mendelsohn, 2009) or calibrated theoretical models (Newell and Pizer, 2003; Fell, MacKenzie and Pizer, 2012).

Our paper also contributes to the empirical literature on measuring the economic effects of environmental regulations.² A number of papers focus on productivity impacts (see, e.g.,

¹For the large literature related to Weitzman’s framework, see, for example, Hepburn (2006), Weitzman (2020), and Stavins and Wagner (2022).

²More broadly, we note that there is a growing literature on the empirical evaluation of industrial policy; see, among others, Kline and Moretti (2014), Aghion et al. (2015), Alder, Shao and Zilibotti (2016), Juhász (2018), Kalouptsidi (2018), Lane (2025), Miravete, Moral and Thurk (2018), Lashkaripour and Lugovskyy (2023), Criscuolo et al. (2019), Rotemberg (2019), Barwick, Kalouptsidi and Zahur (2025), Giorcelli (2019), Yi, Lawell and Thome (2019), Hanlon (2020), Bai et al. (2025), Fan and Zou (2021), Giorcelli and Li (2026),

Jaffe et al., 1995; Berman and Bui, 2001; Greenstone, 2002; Greenstone, List and Syverson, 2012; He, Wang and Zhang, 2020). Using estimated equilibrium models, Ryan (2012) and Fowlie, Reguant and Ryan (2016) emphasize the roles of market power. Chen et al. (2025) study how conglomerates reallocate internal production. In this paper, we develop and estimate a new competitive equilibrium model for China’s coal sector.

We also note that there have been few equilibrium analyses of China’s coal industry in the economics literature despite its enormous importance to China’s economy and the global environment. One reason might be the lack of granular and recent data.³ For example, Zhou et al. (2019) study the evolution of coal mining firms’ productivity, and Zheng (2024) studies the effects of buyer market power on productivity and safety, both using a production function approach based on manufacturing census data before 2007. In our paper, we combine a detailed dataset, institutional knowledge of the market, and the industry reform to address the endogeneity problems in the estimation of coal demand and supply in our equilibrium model. Our estimates of demand and supply elasticities may be of independent interest for other researchers studying China’s coal industry.

In the rest of the paper, we first describe the main institutional features of China’s coal market in Section 2. Then in Section 3, we illustrate the welfare comparison between a tax and a production restriction in a conceptual model. Section 4 describes the various datasets used in the analysis. Sections 5 and 6 present the demand and supply models and their estimation. We use the model to analyze the reform of China’s coal industry and how the observed policy choice reveals policy preferences in Section 7. Section 8 concludes.

2 Empirical Background

2.1 Coal Types, Quality, and Mining Technology in China⁴

Bituminous coal accounts for more than 70% of Chinese coal production and anthracite for another 23%.⁵ High-purity bituminous coal and roughly half of anthracite are used for

Guo and Xiao (2023), Aldy, Gerarden and Sweeney (2023).

³Existing studies that estimate or calibrate the coal demand or supply typically use aggregate annual data (see, for example, Burke and Liao, 2015; Shi, Rioux and Galkin, 2018; Teng, Burke and Liao, 2019). A number of papers have studied the coal-mining industry in other countries, with a focus on safety and productivity (Sider, 1983; Gowrisankaran et al., 2015), technology adoption (Rubens, 2022), labor market competition (Delabastita and Rubens, 2025; Demirer and Rubens, 2025), and the decline of the US coal industry (Watson, Lange and Linn, 2023).

⁴This section is based on Aden, Fridley and Zheng (2009) and National Energy Report (2020).

⁵Coal is usually classified by carbon content measured by weight. The carbon content is over 86% for anthracite and between 45% and 86% for bituminous coal. The remaining coal falls into lower-carbon

coking, which produces coke for steelmaking. The remaining output of these two coal types is used mainly by power plants for electricity production. Lower-purity coal is used primarily in cement production and heating.

Coal purity and extraction methods differ markedly across regions. Shanxi and Inner Mongolia have the largest reserves and are the largest coal-producing provinces, together accounting for more than a quarter of national output.⁶ Shanxi also produces the highest-quality coal, with an average heat content, defined as the total energy released per kilogram after complete combustion, of about 6,242 kcal/kg, compared with a national average of 5,350 kcal/kg and a US average of 5,600 kcal/kg. More than 90% of Chinese mines are pithead mines, where coal is extracted from deep underground. The average mine depth is 456 meters. This extraction method is more costly than open-pit mining, which is more common in countries such as Australia and India and in regions such as the western United States.

2.2 The Formation of the Coal Market

For decades, the Chinese government tightly controlled both coal supply and coal demand through planned prices and quantities. A partial liberalization in 1993 allowed coal to be traded, but preserved special contracts under which large state-owned power plants could buy planned quantities of coal at planned prices (Yang et al., 2018). These special contract prices were typically below prevailing market prices. When contracted volumes fell short of electricity demand, power plants purchased the balance on the spot market. Subsequent reforms culminated in the abolition of the special contracts in 2013 (State Council, 2013). These reforms reshaped the procurement, operating, and investment decisions of power plants, which began to respond to coal-price fluctuations more like profit-maximizing firms (Xu and Chen, 2006; Ma, 2011; Liu, Margaritis and Zhang, 2013; Zhao and Ma, 2013; Gao and Van Biesebeek, 2014; Eisenberg, 2019, 2024).

2.2.1 Method of Sales

Delivery contracts. During our sample period, coal buyers and producers transact through both long-term contracts (annual or multi-year agreements) and the spot market. The major buyers are coal-fired power plants, municipal central-heating systems,⁷ construction firms,

categories, such as subbituminous coal and lignite (American Geosciences Institute, n.d.).

⁶Appendix Figure J.1 shows the distribution of coal reserves.

⁷Across northern China, most residential apartments, government facilities, and schools receive heating through municipal centralized systems that primarily burn coal to generate heat. This infrastructure consists

steelmakers, and fertilizer manufacturers. A long-term contract is signed directly between a coal producer and an end user, rather than a trading intermediary. Provincial governments frequently help broker these matches, particularly in large coal-producing provinces, which supply power plants and heating systems elsewhere in the country (NDRC, 2021*b*). The contracts nominally specify an annual delivery quantity but allow prices to adjust quarterly, or even monthly, in line with spot market prices (e.g., Shanghai Securities News, 2014, 2015; ICBC, 2016). Realized deliveries can therefore depart from stated contract terms (International Energy Network, 2016*a*).

Monthly price discovery. In practice, buyers and sellers in long-term contracts set prices using public monthly price indices. The most widely used indices are regional average transaction prices, net of transportation costs, for coal sold to power plants (NDRC, 2016*c*, 2017). Contract prices are updated using both the initial signing price, which can reflect coal quality and projected supply and demand, and the previous month’s average delivery prices published by coal trade associations and port authorities, which may pool both contract and spot transactions.⁸

Contract compliance. Contemporaneous press reports indicate that almost no contracts were honored at the contracted price between 2014 and 2016. Coal buyers turned to the spot market and stopped buying the contracted coal as soon as the spot market prices fell below the contract level in 2014 and 2015 (Li, 2014; Zhao, 2014, 2015; Bie, 2015; Wang, 2015).⁹ As prices started to surge in 2016, coal producers prioritized selling at high prices in the spot market (Zhu, 2021).

of large coal-fired boilers that heat water, which is then distributed through an extensive network of insulated pipes to buildings (Chen et al., 2013).

⁸International Energy Network (2016*b*) provides an example of how to reset the monthly price using the average price of coal sold to power plants (BSPI) and the average price of coal for other uses (OPI) around the Bohai Gulf, both published by Qinhuangdao Ocean Shipping Coal Trading Market Co. Given the contract price p , the indices p_t^{BSPI} and p_t^{OPI} in month t , and the heat content of coal h measured in kcal/kg, the new price in month $t + 1$ is adjusted to

$$0.5p + (0.25p_t^{\text{BSPI}} + 0.25p_t^{\text{OPI}}) \frac{h}{5,500\text{kcal/kg}}.$$

The government advises buyers and sellers to explicitly specify this formula (dynamic price updates linear in average coal prices) in their long-term contracts. Contract prices, price indices, and assigned weights could differ across firms and years.

⁹Hu (2014) reports that contracts signed at the beginning of 2014 covered 1.9 billion tons of coal, but they were sold at the contracted price only in the first two months of 2014. In our data, production in the first two months typically accounts for about 15% of annual production. Applying this share to the contracted volume implies that, as a reasonable upper bound estimate, about $0.15 \times 1.9 = 0.285$ billion tons were contract-compliant. Relative to 2014’s total production of 3.87 billion tons in our data, this corresponds to a contract-compliant share of about 7.4%.

Compliance improved markedly in 2017 for two reasons. First, new contracts specified a price range that was easier to honor amid spot-market price fluctuations. Second, the government introduced penalties for power plants that failed to buy their contracted volume, including exclusion from the wholesale electricity market (Gao, 2017). These measures compressed the spread between spot and contract prices (China National Coal Association, 2018). Later reports (Zhang, Hong and Wu, 2020) estimate the post-reform annual contract-compliant quantity to be an average of 2 billion tons, roughly 57% of the 2017 quantity.

2.3 Economic Policies

2.3.1 Coal Supply Reform

The 2016 five-year plan from the National Development and Reform Commission (NDRC), China’s macroeconomic management agency, set out a broad agenda for restructuring the coal-mining industry. The plan explicitly called for raising the industry’s profitability, limiting coal production,¹⁰ and reducing the industry’s environmental impact.¹¹ Based on this plan, the State Council, China’s chief administrative body, ordered in February 2016 that coal mines reduce their operating schedules and close small mines.

Implementation timing and compliance varied across provinces and coal mine ownership types. Shanxi, historically the largest coal-producing province, moved first, in April 2016. Other leading coal-producing provinces such as Shaanxi and Inner Mongolia followed in May. Large state-owned mines faced more stringent monitoring and cut output by more (Appendix J.2). Coal prices surged in the second half of 2016, both relative to earlier in the year and relative to the same months of 2015. Strict production monitoring continued through 2017 and 2018. The policy was eased only gradually beginning in 2019 (NDRC, 2019).

2.3.2 Resource Taxation

China introduced a resource tax on coal production in 2014. The central government set a rate band of 2 to 10% on the sales revenues of extracted raw coal,¹² within which provincial governments chose specific rates subject to the central government’s approval (STA, 2014). Major provinces such as Shanxi set the rate at 8%, while others chose lower rates (Qin,

¹⁰In Appendix A, we provide snapshots of the original document and its translation.

¹¹The 13th Five-Year Plan for Coal Industry Development (NDRC, 2016*a*) called for “combining green development with clean utilization to promote green growth” and for “minimizing the ecological and environmental impacts of coal development and utilization.”

¹²This revenue levy does not distinguish among coal grades, but allows for reduced rates on coal extracted from mines near depletion.

Zhang and Xie, 2020). Given this background, we think that a tax-based intervention was an administratively plausible policy alternative to the 2016 production restriction.

Policymakers did not make taxation the principal instrument of the 2016 reform for several reasons. First, the 2014 tax reform was designed to reduce firms’ overall tax burden and fund provincial spending on coal-related environmental damage as part of a broader fiscal reform (Xinhua, 2014), replacing a patchwork of fees on coal production with a single levy. Raising the tax rate to restrict output would have worked against that purpose. Second, fiscal reform moves slowly, whereas the central government viewed cutting coal capacity and output as urgent and better suited to fast, targeted administrative mandates (Li, 2016; People’s Daily, 2016). Third, provincial governments retained resource-tax revenue as an incentive to invest in environmental protection (Hu, 2017), leaving little scope to use the proceeds for redistribution to coal consumers or producers.

2.3.3 Emissions Trading System

China’s carbon market developed over a much longer horizon and was less directly suited to regulating coal production in 2016. Seven regional emissions-trading pilots launched in 2013–14 and, after several years of operation, covered only about 7% of national CO₂ emissions (Almond and Zhang, 2021). The national Emissions Trading System (ETS) did not begin trading until 2021 and initially applied only to the power sector (Goulder et al., 2022). Moreover, the national ETS adopts a tradable performance standard rather than a cap-and-trade design (Ministry of Ecology and Environment, 2020; Goulder et al., forthcoming): allowances are allocated through fuel- and technology-specific emissions-intensity benchmarks, with pre-allocation tied to lagged generation and final allocations adjusted to match realized output.¹³ These features primarily incentivize power plants to reduce emissions intensity rather than to reduce generation, retire coal capacity, or directly restrict coal output. Given its protracted rollout, downstream sectoral coverage, and intensity-based design, we do not view the ETS as a plausible policy alternative to the production restriction in the 2016 reform.

2.3.4 Other Economic Policies

Two further sets of policies during our sample period are worth noting. First, the government expanded credit and infrastructure spending to offset the decline in housing and stock

¹³Allowances in 2019–2020 were first pre-allocated using 70% of 2018 electricity/heat supply, but after 2019–2020 emissions data were verified, the final quota was set according to the generator’s actual 2019–2020 electricity/heat supply. If the final approved quota differed from the pre-allocation, excess allowances were withdrawn or missing allowances were added.

market prices in 2015 and 2016 (Brandt et al., 2020). Many infrastructure projects, such as the national high-speed rail system, expanded rapidly in subsequent years (Reuters, 2016). Second, the government began closing small or unproductive plants in other industries.¹⁴ These closures cut state-owned steel capacity by 10% in 2016 (Lu, 2016). However, we expect the plant closures to have only a small effect on coal demand given that the output share of closed plants is likely low. Between 2016 and 2018, coal demand remained strong as the main consuming sectors—thermal power, coking, steel, and cement—either expanded output or maintained historically high production levels, against the backdrop of continued economic growth.¹⁵

3 Conceptual Framework

We first compare a tax and a production restriction using a local approximation approach based on Weitzman (1974). We model a production restriction as an increase in the slope of supply, because a policy like the one in Section 2 removes production capacity, thus limiting the response of supply to prices. We then show how policy preferences change the comparison. Finally, we develop a complementary global approach to compare the welfare effects of the two policies, applicable to the equilibrium model in our empirical analysis.

3.1 Setup

We specify social welfare from a regulated activity with output q as

$$W(q) = U(q) - c(q) - L(q) \quad (1)$$

where U , c , and L are the gross consumer utility, private producer cost, and pollution damage. The cost of abating the quantity a relative to an output level q is the loss of economic surplus, $C(a; q) = U(q) - c(q) - (U(q - a) - c(q - a))$. Therefore, the marginal abatement cost at q is $C'(0; q) = U'(q) - c'(q)$. The slope of the marginal abatement cost is $C''(0; q) = c''(q) - U''(q)$. Similarly, we use $B(a; q) = L(q) - L(q - a)$ to denote the benefit of abatement, which implies that, at quantity q , the marginal benefit of abatement is $B'(0; q) = L'(q)$, and the slope is $B''(0; q) = -L''(q)$. When both B and C are certain, the optimal quantity $\hat{q}$ satisfies

¹⁴It is separately interesting that shutdowns required government orders rather than occurring through market exit. A plausible explanation is that many plants are supported by local governments to preserve employment despite being unprofitable. To facilitate the shutdown of coal and steel plants, China's Ministry of Finance created a fund of RMB 100 billion (\$15.3 billion) to provide relief to laid-off employees (Lu, 2016).

¹⁵We plot the output of these downstream sectors in Appendix C.1.

$B'(0; \hat{q}) = C'(0; \hat{q})$. With a slight abuse of notation, we write $B'(q)$ and $B''(q)$ for $B'(0; q)$ and $B''(0; q)$ for the rest of the section. We use the same notation for $C'(q)$ and $C''(q)$.

To analyze the welfare effects of policy instruments that shift quantities near $\hat{q}$ under uncertainty, we follow Weitzman (1974) and use a quadratic approximation. Specifically, let $x \equiv q - \hat{q}$ denote the deviation from $\hat{q}$. We assume that there are small, mean-zero, and independent shocks α and β to the marginal social cost and benefit at $\hat{q}$, with variances σ_α^2 and σ_β^2 , respectively. In our context, α captures the shocks to demand and supply, and β captures the shocks to marginal damage from pollution. The welfare difference between q and $\hat{q}$ is approximately

$$\Delta W \equiv W(q) - W(\hat{q}) \approx (\beta - \alpha)x - \frac{C''(\hat{q}) - B''(\hat{q})}{2} x^2. \quad (2)$$

We assume that the marginal benefit of abatement diminishes ($B''(q) < 0$) and the marginal abatement cost increases ($C''(q) > 0$).

3.2 Policy Instruments

The government chooses an instrument before shocks are realized to maximize expected welfare $E[\Delta W]$. The instruments differ in how quantity adjusts after shocks.

Tax (Weitzman, 1974). The regulator sets a per-unit tax on output that equates the marginal social benefit and cost when there are no shocks: $C'(\hat{q}) = B'(\hat{q})$. Given the realized shock α , quantity adjusts so that the private marginal cost matches the new price level, and the realized deviation of quantity from $\hat{q}$ is:

$$x_{\text{tax}} = -\alpha / C''(\hat{q}). \quad (3)$$

Quota. The regulator fixes output at $\bar{q} = \hat{q}$: $x_{\text{quota}} = 0$.

Local production restriction. To make the analysis comparable, we consider a local production restriction that modifies the private cost c in a neighborhood of $\hat{q}$ in two ways. First, the restriction raises the marginal private cost by $L'(\hat{q})$ at $\hat{q}$. This allows the market equilibrium quantity, which is determined by the private demand and supply, to equal $\hat{q}$ in the absence of shocks.¹⁶ Second, the slope of the supply curve is increased by an exogenous

¹⁶Specifically, when there are no shocks, the market equilibrium quantity without any policy is determined by $U' - c' = 0$. When $c'(q)$ is increased by the amount of $L'(\hat{q})$, the modified first-order condition becomes

amount $\kappa \geq 0$. Intuitively, the restriction constrains low-cost capacity and steepens the supply curve, limiting the response of output to cost shocks. Therefore, both c'' and C'' increase by κ . The timing is as follows: the government commits to imposing the restriction that modifies the private cost, then shocks are realized, and the private demand and supply determine the equilibrium quantity. The equilibrium quantity deviation from $\hat{q}$ is (Appendix E):

$$x_{\text{restriction}} = -\frac{\alpha}{C''(\hat{q}) + \kappa}. \quad (4)$$

In the above, the independent shock β associated with B does not affect the equilibrium adjustment between private demand and supply. As a result, the quantity deviation from $\hat{q}$ depends only on α . The restriction can also be viewed as an intermediate case between a tax and a quota: when $\kappa = 0$, it is equivalent to a tax; as $\kappa \rightarrow \infty$, the supply curve becomes vertical and the restriction converges to a quota.

This formulation represents an efficient production restriction. It reduces output according to production efficiency by raising the costs of marginal units of production starting near $\hat{q}$. These are the highest-private-cost units, and the private costs of inframarginal units remain unchanged. A production restriction in the real world can also affect the private costs of inframarginal units, a case we explore in the equilibrium model in our empirical analysis.

3.3 Welfare Ranking

Substituting each instrument's x into (2) and taking expectations yield the following comparisons.

Tax vs. quota (Weitzman, 1974).

$$E[W_{\text{tax}} - W_{\text{quota}}] = \frac{C''(\hat{q}) + B''(\hat{q})}{2(C''(\hat{q}))^2} \sigma_{\alpha}^2. \quad (5)$$

A tax dominates when the marginal social cost increases steeply relative to the rate at which the marginal benefit diminishes ($C''(\hat{q}) + B''(\hat{q}) > 0$); a quota dominates when the marginal benefit diminishes sufficiently steeply ($C''(\hat{q}) + B''(\hat{q}) < 0$).

$C'(q) - L'(\hat{q}) = 0$, or $C'(q) = B'(\hat{q})$. Therefore the equilibrium quantity q would be $\hat{q}$.

Tax vs. local production restriction.

$$E[W_{\text{tax}} - W_{\text{restriction}}] = \underbrace{\frac{C'' + B''}{2(C'')^2} \sigma_\alpha^2}_{\text{tax vs. quota}} - \underbrace{\frac{C'' + B'' + \kappa}{2(C'' + \kappa)^2} \sigma_\alpha^2}_{\text{restriction vs. quota}}, \quad (6)$$

where B'' and C'' are evaluated at $\hat{q}$. We derive this equation in Appendix E. Equation (6) decomposes into the Weitzman difference between tax and quota ($E[W_{\text{tax}} - W_{\text{quota}}]$) minus the welfare difference between the restriction and the quota ($E[W_{\text{restriction}} - W_{\text{quota}}]$). The sign of (6) is ambiguous. When $C'' + B'' < 0$, which is the Weitzman condition under which a quota dominates a tax, the restriction dominates the tax for all $\kappa > 0$ because the steeper supply curve further dampens output variance. When $C'' + B'' > 0$, a small κ can still make the restriction dominate through the variance reduction, but a sufficiently large κ makes the restriction too rigid and the tax dominates.

First- and second-order effects of production restriction. Equation (6) characterizes the second-order local effect of production restriction. Production restriction can also have a first-order effect on welfare by shifting a significant portion of the supply curve, which misallocates production and creates large inframarginal losses, as described in Section 3.2. How the restriction changes the supply curve is an empirical question. In our empirical analysis, data help to identify the shape of the post-policy supply curve, which determines the extent of misallocation. We also use the estimated demand, supply, and pollution damage to construct welfare measures, thus accounting for both the first- and second-order effects.

3.4 Welfare Weights

We now consider a welfare function that weights its components differently. We specialize the analysis to the empirical context in Section 2, where the government prioritizes improving producer surplus and reducing pollution. We focus on comparing a tax paid by coal producers and a production restriction.

In particular, we introduce a weight w_{ps} on producer surplus and a weight w_{damage} on pollution damages. These weights provide a reduced-form way to capture the policy considerations discussed in Section 2.3.2. When tax revenues cannot be effectively redistributed to market participants because of political challenges and administrative costs,¹⁷ one dollar of tax revenue may be worth less to the government's policy objective than one dollar of

¹⁷For example, redistribution may require identifying which firms need support, determining transfer amounts, and verifying that funds are used as intended.

producer surplus. We capture these frictions by allowing $w_{\text{PS}} \geq 1$. Similarly, $w_{\text{damage}} \geq 1$ allows pollution reduction to receive additional policy weight beyond its monetized damage estimate.

Under a tax t , the tax revenue is tq . At the equilibrium price p , the weighted social welfare function is

$$\tilde{W}_{\text{tax}} = U - pq + w_{\text{PS}}(pq - c - tq) + tq - w_{\text{damage}} L. \quad (7)$$

The welfare under a production restriction that generates no tax revenue is:

$$\tilde{W}_{\text{restriction}} = U - pq + w_{\text{PS}}(pq - c) - w_{\text{damage}} L. \quad (8)$$

We now sign how this difference changes as w_{PS} or w_{damage} rises. We assume that the tax maximizes $\tilde{W}_{\text{tax}}$.

Producer weight. Holding $w_{\text{damage}} = 1$,

$$\frac{\partial E[\tilde{W}_{\text{tax}} - \tilde{W}_{\text{restriction}}]}{\partial w_{\text{PS}}} = E[(pq - c - tq)_{\text{tax}} - (pq - c)_{\text{restriction}}] \approx -t\hat{q} < 0, \quad (9)$$

where we ignore the term proportional to the variance of α , which arises from the local approximation and is negligible compared with $t\hat{q}$.¹⁸ The derivative shows that a production restriction is more likely to be the better instrument when the government values producer surplus. The intuition is that the amount $t\hat{q}$ extracted from producer surplus (weighted by $w_{\text{PS}} > 1$) is devalued when it is converted to tax revenue (weighted by one).

Damage weight. We calculate the derivative with respect to w_{damage} while holding $w_{\text{PS}} = 1$:

$$\frac{\partial E[\tilde{W}_{\text{tax}} - \tilde{W}_{\text{restriction}}]}{\partial w_{\text{damage}}} = -\frac{L''}{2} \left(\frac{\sigma_{\alpha}^2}{(C'')^2} - \frac{\sigma_{\alpha}^2}{(C'' + \kappa)^2} \right). \quad (10)$$

We derive this equation in Appendix E. The sign depends on the curvature of the damage function. When $L'' > 0$ and $\kappa > 0$, the derivative is negative: a greater weight on pollution damage favors the restriction because its steeper supply curve generates less output variance and hence less expected damage when damage is convex. We note that this result highlights the second-order effects of policies near $\hat{q}$. In reality, a production restriction may mistarget $\hat{q}$ and can reduce a quantity less than a tax, causing a higher w_{damage} to favor the tax and potentially offsetting the second-order effect.

¹⁸The term we ignore in the derivative is equal to the right-hand side of (5).

3.5 Comparing Instruments Using Stochastic Dominance

The welfare analysis above delivers tractable analytical results, but it relies on a local approximation, requires curvature assumptions on benefits and costs, and ignores the impact of a production restriction on the inframarginal supply curve. We now present a complementary, global comparison of a tax and a production restriction. We first present the criterion in a setting where the researcher can compute the equilibrium distributions of output and economic surplus under each policy, for example, using estimated demand and supply functions, but does not know the curvature of the pollution damage function L . Then we consider a dual formulation when the damage is known but the economic surplus is not.

The key idea is to use stochastic dominance to identify when a tax improves upon a production restriction for any increasing damage function. A classical result in decision theory (Quirk and Saposnik, 1962; Hadar and Russell, 1969) provides a tool for comparing $E(L(Q))$ across policies without knowing the curvature of L : for any increasing function L and two random variables Q_1 and Q_2 , $E(L(Q_1)) \leq E(L(Q_2))$ if their distributions satisfy $F_{Q_1}(q) \geq F_{Q_2}(q)$ for all q (first-order stochastic dominance). Here we use the upper case Q to emphasize that the equilibrium quantity is a random variable driven by uncertainty in demand and supply, and the expectation $E(L(Q))$ is taken over the resulting distribution of Q . The framework can additionally accommodate a vector of random variables ρ , representing uncertainties unknown ex ante to the government, that also affects the damage function, $L(Q; \rho)$. We assume that ρ does not affect the distribution of Q , and the expectation $E(L(Q; \rho))$ is over both Q and ρ . The procedure below is still valid if $L(Q; \rho)$ is an increasing function of Q for each possible value of ρ .

The procedure assesses whether a tax can simultaneously improve upon the expected economic surplus and the avoided damage implied by a production restriction. Given the expected social welfare:

$$W^* = \underbrace{E(U - c)}_{\mathcal{E}} - E(L(Q)),$$

1. compute the expected economic surpluses $\mathcal{E}_{\text{restriction}}$ and $\mathcal{E}_{\text{tax}}$ under the production restriction and the tax;
2. compute the respective distributions of quantities;
3. if (a) the distribution of output under the tax is stochastically dominated by that under the production restriction, which implies $E(L(Q_{\text{tax}})) \leq E(L(Q_{\text{restriction}}))$ for any increasing L , and (b) $\mathcal{E}_{\text{tax}} \geq \mathcal{E}_{\text{restriction}}$, the tax achieves higher W^* .

We note that the procedure above relies on a sufficient condition to compare a tax and a production restriction. When one of the two conditions in step 3 fails, the procedure does not predict which instrument achieves higher welfare.

More broadly, this procedure is valid for comparing any two policies: they do not need to maximize social welfare, and they may be subject to constraints outside our model. The comparison can be used to determine which policy is relatively more efficient as long as the two conditions in step 3 hold.

Why a surplus-improving tax can have lower output. A production restriction that constrains low-cost producers makes the aggregate supply curve steeper. A tax, by contrast, preserves the original supply curve and reallocates production toward lower-cost producers. Figure 1 illustrates this point: at any given level of aggregate output, a tax generates more economic surplus (the shaded area between demand and supply) than a production restriction with a steeper supply $S_{\text{restriction}}$. Equivalently, to achieve the same surplus, the equilibrium quantity under a tax (Q_{tax}) is lower than under the restriction ($Q_{\text{restriction}}$). When the tax can generate at least as much surplus with lower quantities across the support of the shocks, the output distribution under the tax is stochastically dominated, and the tax improves welfare for any increasing damage function. This is a result of the first-order effect described in Section 3.3.

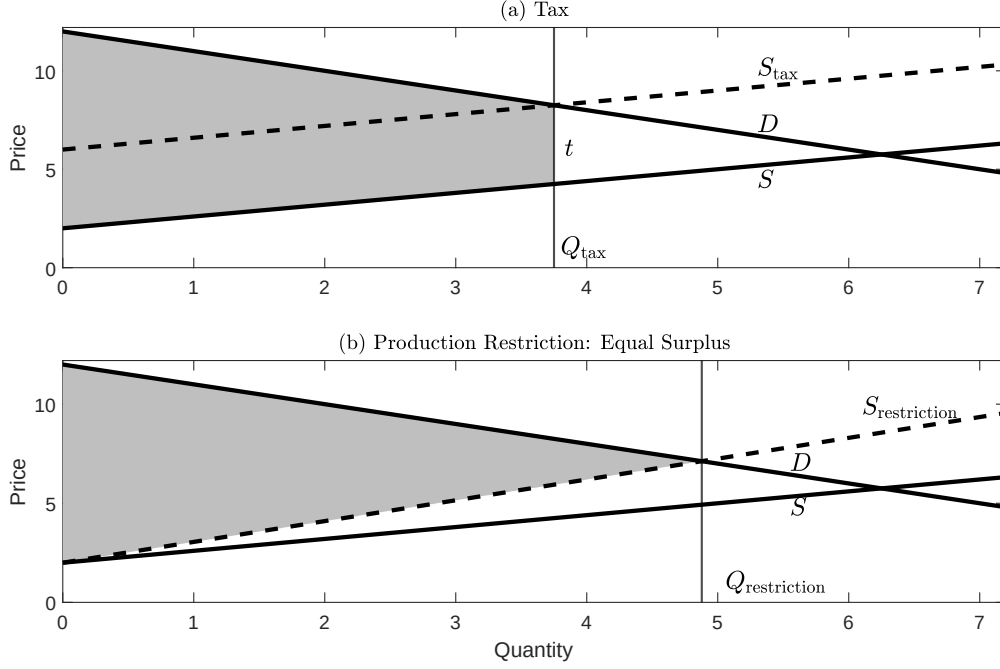
A dual formulation. The same logic extends to comparing policies when the benefits of abatement are relatively well measured but the costs of abatement are not. Suppose a pollution-control policy reduces market quantity below the no-policy equilibrium. Under the condition that private surplus (net of cost) increases in quantity up to the no-policy equilibrium level, then between two policies that reduce pollution damage by the same amount, the policy that induces a stochastically dominating distribution of market quantity, thus preserving higher output, delivers higher welfare.

4 Data

4.1 Data Sources

Our primary dataset comes from the China Coal Transportation and Distribution Association (CCTDA), a major trade group in the coal industry. The data include monthly coal shipment quantities between provinces from 2012 to 2016. The tonnage data combine raw coal, washed

Figure 1: Tax Generates More Surplus at Lower Quantity than Production Restriction



Notes: Panel (a): under a tax t , the equilibrium quantity is Q_{tax} and the shaded area is the economic surplus. Panel (b): a production restriction rotates supply from S to $S_{\text{restriction}}$; to generate the same surplus, the quantity $Q_{\text{restriction}}$ must be higher.

coal, and other derivative products with high coal content, such as coke and briquettes.¹⁹ The shipment data include shipments both within a province and to other provinces.

We then compile additional data from other sources. We collect province-level mining capacity data for 2014–2018 from the National Energy Administration and the National Mine Safety Administration. Capacity changes are approved every 6 months.²⁰ We also obtain province-month production data for 2014–17 from CCTDA and for 2018–2019 from the National Bureau of Statistics (NBS). The production data reflect the total tonnage of excavated raw coal. We obtain from CCTDA monthly production data for key state-owned (SOE) coal mines, which are the largest coal mines in each province.²¹ Monthly coal imports,

¹⁹Consistent with the standard NBS practice, the data account for transportation of final products from raw coal producers to end users. More specifically, if 1 ton of raw coal was extracted in province A , transported to province B for washing (removing impurities), with quantity reduced to $x < 1$, and burned at power plants in province C , the data would record only x as transported from A to C and would not count the A – B transportation. We also note that not all raw coal needs washing, and many coal mines are integrated with washing facilities and need not transport coal to a third province for processing. Our data closely match the NBS aggregate production and consumption statistics. See Appendix B for more details.

²⁰In principle, annual production should not exceed the approved level, and monthly production should not exceed 1/12 of the annual level by more than 10%.

²¹NDRC designates an SOE coal mining firm as a “key SOE” based on a number of criteria, which include capacity, reserves, mining technology, and coal quality (whether the coal is mainly for electricity production).

measured in total tonnage, from outside China to each province between 2014 and 2017 are from the firm International Coal.²² We collect monthly provincial coal prices for 2014–2019 from NDRC.²³ To study the impact of coal consumption on air pollution, we collect air quality data, including hourly concentration measures of pollutants such as PM2.5 and PM10, at the monitoring-station level from the Ministry of Environmental Protection.

We do not directly observe monthly province-level coal consumption. To construct coal consumption at the province-month level, we impute the quantities based on (1) shipment data from CCTDA, (2) monthly output of major coal-using industries, and (3) annual province-level coal consumption, with the latter two from NBS. Details of this imputation are in Appendix C.

To verify the consistency of data from different sources, we compare CCTDA’s outbound shipment data with the NBS production data in Appendix B. Our shipment data are slightly lower, by about 4% to 8%, than the NBS statistics. The same pattern holds consistently across provinces.

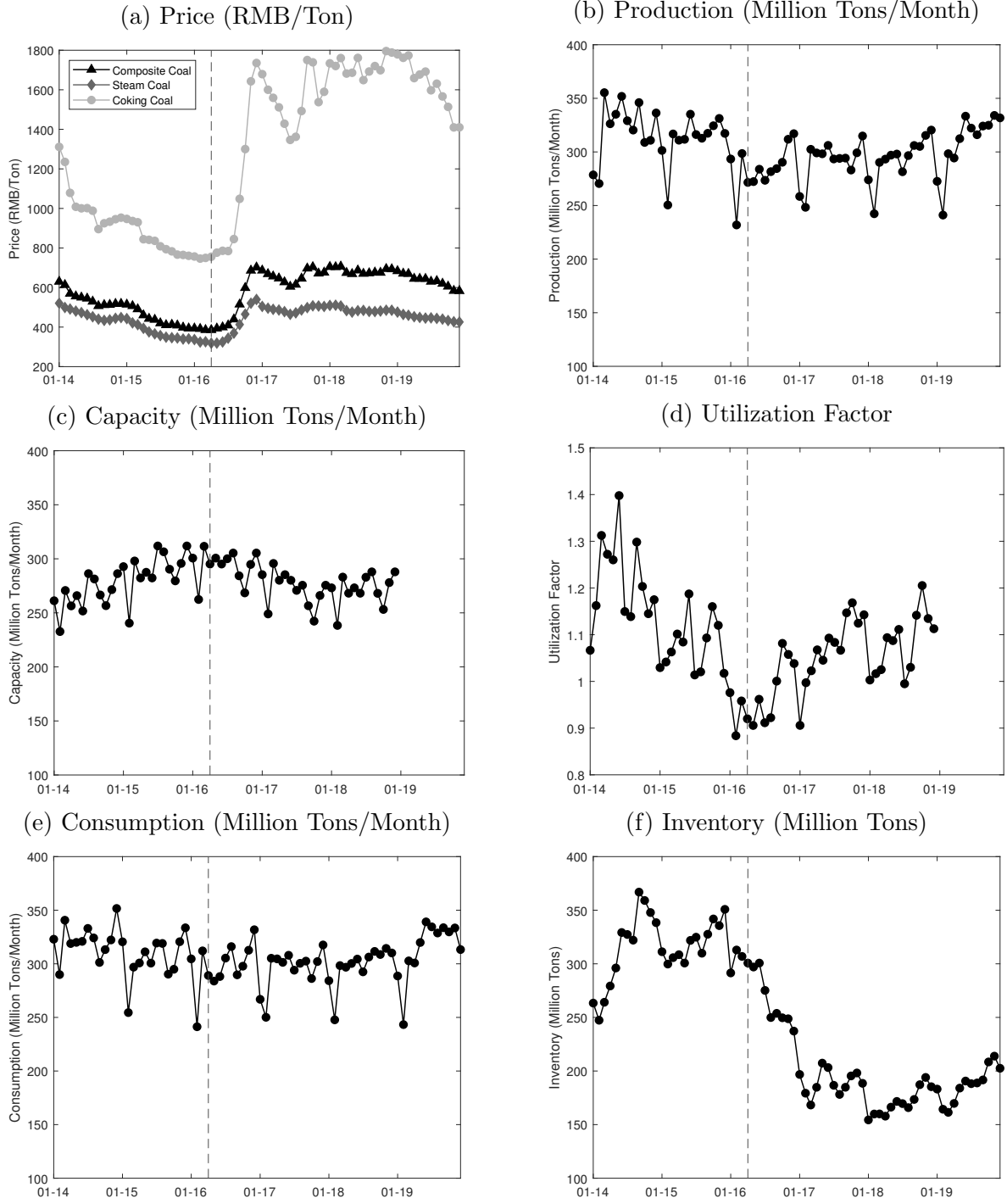
Data Aggregation

In the rest of the paper, we use the total tonnage of coal produced and consumed to analyze supply and demand. We do not further differentiate among coal types within the quantity data. Given our research questions, several potential concerns about the data warrant discussion. In particular, to what extent can a single demand function capture the composite demand for the variety of coal and coal products combined in our quantity data? What is the appropriate price for this composite demand? We note that, in China, demand for different coal products is likely highly correlated. The widespread use of coal-blending techniques enables power plants to substitute across different coal types in the event of shortages. Other end users, such as heating facilities, are likely even less sensitive to coal quality. As we show next, thermal and coking coal prices are highly correlated. We therefore construct an average price for each province and month based on the two price series, weighted by their consumption shares.

²²The firm provides the data through the website <https://mcoal.in-en.com/>. We do not consider China’s exports, which account for less than 0.2% of coal production during our sample period.

²³The publicly available coal price data are based on coal used for electricity generation and heating. This type of coal, which includes a variety of coals with different heat contents and purity, is collectively referred to as “steam coal” or “thermal coal.” We collect monthly province-specific prices of steam coal using the China Steam Coal Price Index from the Price Monitoring Center of NDRC. The index is based on data collected from 1,600 firms that rely on coal as the main fuel source, including major coal-fired power plants and heating facilities. Prices of coal used for coking are based on prices in Taiyuan in Shanxi, a major source of coking coal in China.

Figure 2: Time-Series Data of the Coal Market



Notes: Panel (a) plots the prices of steam coal, coking coal, and the composite price; Panel (b) plots national monthly production; Panel (c) plots monthly national capacity, adjusted from the six-month capacity data using the number of working days in each month (our capacity data end at the end of 2018); Panel (d) plots the utilization factor, defined as national monthly production divided by capacity; Panel (e) plots national monthly consumption; and Panel (f) plots national inventory. The vertical line indicates the start of the reform.

4.2 Data Summary

4.2.1 National Time Series

We plot the time series of national coal prices, production, capacity, utilization factor, inventory, and consumption in Figure 2. Coal prices in Panel (a) increased significantly for both coking and steam coal after the policy started in Shanxi in April 2016. Prices remained higher throughout 2017 and 2018 and gradually declined in 2019 after policy pressure eased. Panel (b) plots monthly production. Production in 2016 was noticeably lower than in similar periods before the policy.

The supply squeeze resulted from two aspects of the policy. First, the government closed coal mines and suspended new approvals, resulting in a net decrease in capacity, as shown in Panel (c). Second, the campaign to reduce overproduction above approved capacity levels directly limited output. In Panel (d), we plot the utilization factor, defined as the production-to-capacity ratio. This ratio in 2016–2018 after the policy is lower than or comparable to that in 2014–15, despite significantly higher prices in the later years.

Panels (e) and (f) show that the market went through a transition in 2016. Panel (e) shows that consumption responded more modestly than production in 2016. Excess demand over production coincided with a more than 40% decrease in inventory during the rapid price increase in 2016 (panel (f)).²⁴ Inventory stabilized at a lower level in 2017 and 2018, when production increased above 300 million tons per month, matching the level of consumption.

²⁴Monthly foreign imports also increased by about 10 million tons per month in 2016 and remained at about 20 million tons per month after the policy.

Table 1: Summary Statistics

	Mean	SD	p25	Median	p75
<i>Panel A. Coal-Consuming Provinces</i>					
Consumption (Million Tons/Month/Province)	10.16	7.63	4.62	7.47	13.30
Price (RMB/Ton)	532.01	133.87	430.21	523.03	615.45
Wind Generation (TWh/Month/Province)	0.53	0.73	0.06	0.29	0.70
Cooling Degree Days ($^{\circ}\text{C}/\text{Month}/\text{Province}$)	17.77	38.16	0	0	11.12
N=1440; based on 30 coal-consuming provinces in 2014–2017					
<i>Panel B. Coal-Producing Provinces</i>					
Quantity (Million Tons/Month/Province)					
Supply	11.30	19.91	1.20	3.61	9.98
Capacity	11.21	17.85	1.59	5.18	12.40
N=1200; based on 25 coal-producing provinces in 2014–2017					
<i>Panel C. Foreign Import</i>					
Quantity (Million Tons/Month)	18.12	4.97	14.00	17.55	21.82
N=48; 2014–2017					

Notes: Panel A shows summary statistics for 30 coal-consuming provinces. The demand consists of shipments from Chinese provinces and foreign imports. Coal consumption is aggregated from consumption in all sectors using coal for each province (Appendix C). The price is the quantity-weighted average of thermal and coking coal prices plus a calibrated transportation cost (Appendix F) based on coal origins. We also use monthly wind generation and cooling degree days in the analysis. Panel B reports summary statistics for 25 coal-producing provinces. Supply is the sum of all shipments from that province each month. Capacity is aggregated from the mine-level data from NEA. Panel C reports total imports.

4.2.2 Summary Statistics

For the main analysis of the paper, we focus on the years from 2014 to 2017.²⁵ Table 1 summarizes our main dataset. There are 30 coal-consuming provinces and 25 coal-producing provinces. All coal-producing provinces also consume coal. The reported prices include transportation costs (based on coal origins), which account for approximately 17.5% of the composite price.²⁶

²⁵CCTDA’s cross-province shipment data are available through 2016, and its production data are available through 2017. We cross-validate both datasets using NBS statistics. For subsequent years, NBS reports province-level monthly production.

²⁶Appendix F details how we calibrate the transportation costs.

Table 2: Mean and Standard Deviation of Cross-province Shipment Shares, 2014–2016

<i>Producing Province</i>	<i>Consuming Province</i>									
	Inner Mongolia	Shanxi	Shandong	Hebei	Jiangsu	Shaanxi	Henan	Liaoning	Guizhou	Xinjiang
Inner Mongolia	0.392 [0.014]	0.011 [0.002]	0.049 [0.008]	0.017 [0.003]	0.064 [0.010]	-	0.002 [<0.001]	0.082 [0.003]	-	-
Shanxi	0.006 [0.002]	0.298 [0.013]	0.104 [0.007]	0.231 [0.029]	0.103 [0.012]	0.002 [0.001]	0.055 [0.012]	0.017 [0.002]	-	0.001 [0.001]
Shaanxi	0.004 [0.001]	0.148 [0.020]	0.086 [0.049]	0.039 [0.010]	0.101 [0.027]	0.411 [0.070]	0.059 [0.007]	-	0.001 [0.001]	-
Guizhou	-	-	-	-	-	-	-	-	0.740 [0.025]	-
Shandong	-	-	0.857 [0.016]	0.012 [0.005]	0.033 [<0.001]	-	0.008 [0.001]	0.002 [<0.001]	-	-

Notes: The table reports the shipment shares of the five largest coal-producing provinces to the ten largest coal-consuming provinces, averaged over 36 months between 2014 and 2016. The share is calculated as a month's shipment from i to j divided by the total shipment from i in the month. The standard deviations are reported in square brackets.

4.2.3 Cross-province Shipment Shares Are Stable Over Time

We show that the share of a province’s coal production transported to another province is highly stable over time. This empirical fact will guide the modeling decisions in the next section. In Table 2, we report the mean and standard deviation (in square brackets) of the shipment shares from province i to province j , defined as $\frac{Q_{ijt}}{Q_{it}}$, where Q_{ijt} is the tonnage of coal produced in i and transported to j , and Q_{it} is the total production in i in month t . We list the five largest coal-producing provinces in rows and the ten largest coal-consuming provinces in columns. The standard deviation for most province pairs is negligible relative to the mean, indicating a high degree of stability in shipment shares over time.

There are two possible reasons for this stability despite substantial variation in coal prices over time. First, the government and railway operators allocate railway capacity when coal buyers and sellers enter into long-term contracts. These rail-capacity allocations create rigidity within the duration of long-term contracts, as any significant rerouting of shipment across provinces could create cascading disruptions in the interconnected rail network. This constraint incentivizes coal buyers to switch between suppliers within a given province, even when they procure coal outside their long-term contracts. Second, although power plants can substitute across different grades of coal to some extent through coal-blending technology, they still tend to prefer similar coals when purchasing on the spot market, because coal blending can raise plants’ operating costs (Zhao, 2021). Since mines within the same province are more likely to produce similar grades of coal, they may supply to the same set of power plants, consistent with stability of shipment shares over multiple years.

We also find support for these structural reasons behind the rigidity of shipment shares. Between 2014 and 2016, the log difference between the highest and lowest coal prices across provinces ranges from 0.62 to 0.94.²⁷ If shipment shares could adjust flexibly in response to price differentials, we would expect to observe strategic shipping behaviors that exploit arbitrage opportunities, keeping price dispersion relatively low. Instead, the observed persistence of price differentials points to a lack of such strategic behavior. Our model in the following sections thus focuses on production decisions rather than shipment-destination choices.

²⁷The log difference between the 25th and 75th percentile prices ranges from 0.12 to 0.35.

5 Coal Demand

We specify coal consumption in province j and month t as

$$\ln q_{jt}(p_{jt}) = \alpha^D \ln p_{jt} + \beta_X X_{jt} + FE_j^{\text{demand}} + FE_{m(t)}^{\text{demand}} + \varepsilon_{jt}, \quad (11)$$

where q_{jt} is coal consumption in province j , and p_{jt} is the composite coal price in the province. The parameter α^D has the straightforward interpretation of a price elasticity. The covariates include province j 's one-month lagged cooling degree days (CDD) and wind generation, which affect the local electricity residual demand for coal power plants, and an indicator for 2017 to capture government intervention to shore up long-term contract compliance on the demand side. We also include fixed effects for province and month of the year. The term ε_{jt} captures the unobserved demand shock.

We treat wind generation as largely exogenous to the unobserved shocks ε_{jt} for two reasons. First, wind power build-out is primarily driven by centrally planned investment that targets areas rich in wind resources (State Council, 2014). Second, NDRC directs local governments and the power grid to guarantee priority dispatch and full purchase of wind generation (NDRC, 2016b). Given these policies, we assume wind production is driven by an exogenous long-term trend plus local and short-run weather conditions, which generates exogenous variations in residual demand for coal generation.

Static demand assumption. In theory, the coal purchase decisions can be dynamic, as coal users can rely on existing inventory to smooth consumption when prices change (Jha, 2023). Our demand function is static, because coal consumers do not store large quantities of coal in our context. Specifically, NDRC requires coal power plants to keep enough stock for at least 7 days of operation, but explicitly caps inventories at no more than 12 days in peak seasons (NDRC, 2021a). A historically high value in 2024 is just 27 days (CCTV News, 2024). Major steel plants keep a similar level of coal inventory (BOCI, 2017). This practice contrasts sharply with the norm in the US, where over 90% of coal power plants keep enough coal to generate electricity for over 60 days (EIA, 2023). We also note that, although there may still be arbitrageurs with sizable capacity, such as dedicated coal traders storing coal at ports, the model in (11) focuses on coal consumption by end users instead of shipment. In our counterfactual analysis, we focus on the “steady state” of the market, where demand matches the level of consumption in the markets before 2016 or in 2017, and inventory levels were roughly stable over time (Figures 2, Panels (e) and (f)).

Long-term contracts. Another concern is that long-term contracts may reduce demand for spot-market coal. Section 2 shows that this is largely not the case for 2014–2016: buyers frequently purchased from the spot market when market prices were low, and sellers sold to the spot market when prices were high. In 2017, government measures both improved contract compliance and brought contract prices more in line with spot prices, reducing the distinction between demand for contracted and spot-market coal. As a first-order approximation, we model coal transactions as occurring in the spot market. We further explore the robustness of our demand estimates in Appendix H, where we assume that contracts reduce demand for spot-market coal.

5.1 Identification

The equilibrium price may be correlated with the unobservable ε_{jt} . We construct shift-share instrumental variables that exploit differences across provinces in the timing of, and exposure to, the production restrictions as documented in Section 2.3.1.

Instrumental Variables (IV)

There are three potentially useful variations:

1. The policy started first in Shanxi in April 2016, followed by other provinces in May. Shanxi is also one of the provinces that experienced the largest decrease in output.
2. The policy imposed stricter monitoring programs on the production of state-owned enterprises (SOEs), and the SOE share of production differs across provinces.
3. The policy closed small coal mines, and the cutoff thresholds differ across provinces.

In each province, variation in policy timing and intensity shifts local supply through the largely fixed shipment shares (Section 4.2.3). Importantly, because the demand function (11) already accounts for policy effects through the 2017 indicator, our excluded instruments are based on purely supply-side variations in 2016, when the policy was rolled out across provinces at different times and with different intensity. Specifically, we use χ_{it} to denote whether the policy applied to province i 's supply in month t . For each province j in month t , we define (1) a policy IV

$$Z_{jt}^{\text{Policy}} = \sum_i w_{ij}^{2012-2013} \chi_{it}, \quad (12)$$

where $w_{ij}^{2012-2013}$ is coal shipped from i to j divided by the total shipment to j , averaged across months in 2012 and 2013, before the estimation sample. This measure captures j 's historical dependence on coal from i .²⁸ We also define (2) an SOE IV,

$$Z_{jt}^{\text{SOE}} = \sum_i o_i w_{ij}^{2012-2013} \chi_{it}, \quad (13)$$

where o_i is the output share²⁹ of key SOEs in province i , also averaged across months in 2012 and 2013. The interactions between o_i and $w_{ij}^{2012-2013}$ proxy for differences in exposure to the supply-side policy shocks. Finally, we define (3) a capacity IV,

$$Z_{jt}^{\text{Capacity}} = \sum_i K_i^{\text{Small}} w_{ij}^{2012-2013} \chi_{it}, \quad (14)$$

and K_i^{Small} is province i 's capacity in January 2014 below the province's cutoff thresholds for closure during the policy in 2016.³⁰

IV Validity

Our IVs exploit the independence between demand shocks and the shipment shares' interaction with χ_{it} , conditional on province fixed effects and month-of-the-year fixed effects. This is the key identifying assumption for the validity of Bartik-type shift-share instruments (Goldsmith-Pinkham, Sorkin and Swift, 2020). This assumption is supported by the policy's implementation pattern: the earliest and largest effects occurred in Shanxi, and SOE production faced tighter restrictions because the government prioritized large coal mines to achieve rapid output reductions (Appendix A), rather than because of correlations with demand shocks. In particular, SOEs in the two largest producing provinces significantly reduced their production in 2016 (Appendix J.2).

One potential concern with the exclusion restriction of the shift-share IVs is a correlation between policy start times and other economic policies. We note that the 2015–2016 economic stimulus plan is a nationwide policy that generally eases the borrowing constraints

²⁸We find that the shares of shipment from a province, w_{ij} , like the shares of shipment to a province in Section 4.2.3, are stable over time during our sample. We argue in Section 4.2.3 that this rigidity is likely due to structural reasons.

²⁹Ideally, we would use capacity shares to construct the IV because there may be serial correlations in unobservables. We do not have capacity data for SOE mines. The main assumption is that any serial correlation of production between 2013 and 2016 is negligible.

³⁰Following the official policy document, the capacity cutoffs for “small” coal mines to be closed are 0.3 million tons/year for major coal-producing provinces: Shanxi, Inner Mongolia, Shaanxi, and Ningxia; 0.15 million tons/year for Hebei, Liaoning, Jilin, Heilongjiang, Jiangsu, Anhui, Shandong, Henan, Gansu, Qinghai, and Xinjiang; and 0.09 million tons/year for the remaining provinces.

for transportation projects, such as the national network of high-speed rail (HSR). A correlation with our instruments may arise if Shanxi, the first province to enact the policy, is also a major supplier to provinces with more infrastructure projects in April that would use more coal. Appendix D lists major active infrastructure construction projects in April and May of 2016. The large volumes of concrete and steel used in these projects could generate substantial demand for coal and coke. We find no clear correlation between the locations of these projects and Shanxi’s main destination provinces—Shanxi itself, Hebei, Shandong, and Jiangsu—which together account for over 70% of Shanxi’s output (Table 2).

Another concern is that policy timing may be endogenous if coal buyers anticipated the policy and stockpiled coal in advance. Policy guidance and common industry practices discussed previously show that major end users of coal do not have significant storage capacity. National coal inventory data from NBS in Figure 2 show that the inventory actually declined before the policy, suggesting that the policy was likely a surprise.

5.2 Estimation

To understand the strengths of the IVs and the underlying variations they capture, we report the first-stage results in Table 3. Columns (1)–(3) regress the log of prices on each instrument in the previous section. We find that Shanxi’s policy implementation raises prices in provinces more exposed to its supply. Similarly, prices are higher in provinces where a greater share of supply historically came from SOEs or small mines and, given the intensity of the policy, particularly Shanxi’s SOEs and small mines. The F statistics indicate that the policy and SOE IVs are considerably stronger.

Table 4 reports the second-stage results based on the instruments in Table 3. Both tables report province-clustered wild-bootstrap standard errors. We use the estimated elasticity of -0.635 based on the policy and SOE IVs in column (5) as our primary specification. Notably, the stronger IVs in columns (2), (3), and (5) all imply similar demand elasticities. Although the estimate based on the weaker small-mine-capacity IV in column (4) is larger, the estimate in our primary specification is not rejected at the 95% confidence level. The IV estimates are considerably different from the positive OLS estimate in column (1). For comparison, prior work (Burke and Liao, 2015) estimates the elasticity for coal demand to be between -0.3 and -0.7.

Table 3: Demand Estimation: First-Stage

	(1) Policy	(2) SOE	(3) Small-Mine Capacity	(4) Policy+SOE
Z_{jt}^{Policy}	0.075*** (0.014)			0.017 (0.030)
Z_{jt}^{SOE}		0.147*** (0.023)		0.117** (0.050)
Z_{jt}^{Capacity}			0.086*** (0.033)	
Wind Generation (TWh)	-0.067*** (0.024)	-0.066*** (0.025)	-0.056*** (0.022)	-0.067*** (0.025)
CDD/1000	0.260*** (0.081)	0.249*** (0.087)	0.335*** (0.101)	0.249*** (0.084)
Province FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
Kleibergen-Paap F Statistic	24.91	37.02	5.93	19.23
N	1440	1440	1440	1440

Notes: The dependent variable is log delivered coal price. Columns show results using different IVs. Province and month fixed effects are included in all specifications. Standard errors, reported in parentheses, are clustered at the province level using a wild bootstrap. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

Table 4: Demand Estimation

	OLS	(2) Policy	(3) SOE	(4) Small-Mine Capacity	(5) Policy+SOE
$\ln p_{jt}$	0.115*** (0.031)	-0.696*** (0.212)	-0.619*** (0.197)	-1.258*** (0.487)	-0.635*** (0.204)
Wind Generation (TWh)	-0.039** (0.016)	-0.081*** (0.027)	-0.077*** (0.024)	-0.110*** (0.040)	-0.078*** (0.023)
CDD/1000	0.465** (0.225)	0.742*** (0.238)	0.716*** (0.228)	0.934*** (0.296)	0.722*** (0.237)
Province FE	YES	YES	YES	YES	YES
Month FE	YES	YES	YES	YES	YES
N	1440	1440	1440	1440	1440

Notes: The dependent variable is log coal consumption. Columns show results using different IVs. Province and month fixed effects are included in all specifications. Standard errors, reported in parentheses, are clustered at the province level using a wild bootstrap. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

6 Coal Supply

Our supply model incorporates two key institutional features of China’s coal industry. First, the coal market is not concentrated. In Appendix J.3, we plot the combined output shares of the three largest coal-mining firms, based on output in the key SOE production data, in each of the top five coal-producing provinces. No single firm accounts for more than 30% of output in a province, and the share of the third-largest firm is often below 10%. Second, as we show in Section 4.2.3, bilateral shipment shares across provinces are highly stable over time.

Motivated by these two observations, we specify the pre-policy aggregate marginal cost function at the province level as

$$\ln mc_{it} = \alpha^S \frac{q_{it}}{K_{it}} + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \quad (15)$$

where K_{it} is capacity in province i in month t , which is taken as given in the data. We include province and month-of-year fixed effects. We assume that, in equilibrium, the marginal cost equals the effective price $P_{it} = \sum_j s_{ij} (p_{jt} - \tau_{ij})$, where s_{ij} is the share of coal shipped from i to j in total shipments from i ,³¹ p_{jt} is the price in the destination province j , and τ_{ij} is the calibrated transportation cost following the procedure in Appendix F.³²

Behavioral foundation. The marginal cost function (15) implies the supply curve $\frac{Q_{it}(P_{it})}{K_{it}} = \gamma \ln P_{it} + \widetilde{FE}_i^{\text{supply}} + \widetilde{FE}_{m(t)}^{\text{supply}} + \tilde{\xi}_{it}$, where $\gamma = \frac{1}{\alpha^S}$, with fixed effects and shocks appropriately transformed. The supply function can be microfounded as the aggregate decisions of many atomistic, competitive mines. Suppose each producer in province i sells a share s_{ij} to province j . The cost function is given by $c(q; \theta)$, where the cost parameter θ differs across mines and follows the distribution Θ_i . The total mass of the mines is $\mathcal{M}$. Slightly abusing notation, we use $o_{\theta t}$ to denote this producer’s output decision, which satisfies the first-order condition $P_{it} = c'(o_{\theta t}; \theta)$. Under the assumption that the marginal cost function c' is strictly monotonic in $o_{\theta t}$ for any θ , we can invert the production decision as $o_{\theta t} = c'^{-1}(P_{it}; \theta)$. Integrating both sides with respect to $\Theta_i(\theta)$ and multiplying the total mass of mines $\mathcal{M}$ yields an aggregate supply curve as a function of P_{it} , and appropriate choices of c and Θ_i can generate the specific exponential form.

A caveat of this microfoundation is that the effective price P_{it} would no longer be a suffi-

³¹We use the average shares as in Table 2, and the results are similar if we use any single year’s data to construct the shares.

³²We also include the resource tax in the effective price.

cient statistic for all relevant prices if the output share s_{ij} is heterogeneous across individual mines. In the extreme case, each mine may supply coal to only one destination province, as opposed to splitting its output across provinces. Although we do not have mine-level data on output and shipment, we can rule out this extreme case based on news reports showing that large mines routinely supply buyers in multiple provinces (Xinhua, 2016; China Securities Journal, 2017; Xinhua, 2018).

Static supply assumption. We assume that the coal output decisions are static. According to our interviews with industry practitioners, mine owners plan production based on current prices, and dynamic factors such as the amount of unexcavated coal or interest rates do not appear to be first-order drivers of their production decisions. For context, coal reserves in China are close to 150 billion tons, while annual production varies between 3 and 4 billion tons, with a similar amount of new reserves discovered each year.

Supply function after policy. Given that we observe only 25 provinces, our model of the policy effects is necessarily parametric. We estimate the following specification, similar to Ryan (2012):

$$\ln mc_{it} = \alpha^S \frac{q_{it}}{K_{it}} + \alpha^{S,\text{policy}} \chi_{it}^{\text{policy}} \left(\frac{q_{it}}{K_{it}} - \nu \right)_+ + FE_i^{\text{supply}} + FE_{m(t)}^{\text{supply}} + \xi_{it}, \quad (16)$$

where $\chi_{it}^{\text{policy}}$ is an indicator for the policy period, beginning in April 2016 for Shanxi and May 2016 for other provinces, and $(x)_+ = \max(x, 0)$. The term $\left(\frac{q_{it}}{K_{it}} - \nu \right)_+$ captures the additional effect of the policy on marginal costs when production approaches capacity, with the cutoff ν estimated from the data. We allow the policy to affect α^S , thereby changing the province-level aggregate marginal cost curve once production exceeds a threshold. This functional form lets the data discipline how well the government can target inefficient coal mines and how sharply marginal costs rise once production exceeds the cutoff. It therefore allows us to quantify the misallocation induced by a command-and-control production restriction. In the extreme case where $\alpha^{S,\text{policy}}$ is large, the policy effectively mimics an efficiently implemented within-province quota, with quota level νK_{it} .³³

³³A potential concern with the specification is that the policy in 2016 initially imposed a 276-working-day restriction that was phased out in December, while the 2017 policy did not. Policies in both years increased monitoring and reduced production by limiting overtime production. With limited data, we pool the periods across the two years to estimate the nonlinear effects of the policy. The data suggest that assuming the same policy effect is a reasonable simplification, because the quantity difference due to the 276-day restriction itself is likely small. In the data, total coal production is 3.410 billion tons in 2016 and 3.491 billion tons in 2017, 2% higher. Over the same months, June to November, when the 276-day restriction is active in 2016 but not in 2017, production levels are also similar, at 1.73 billion tons in 2016 and 1.77 billion tons in 2017.

6.1 Identification

The supply function and the resulting estimation equation can be written in the general form

$$\ln P_{it} = g\left(q_{it}, X_{it}^{\text{supply}}\right) + \xi_{it}, \quad (17)$$

where q_{it} is correlated with ξ_{it} , and X_{it}^{supply} is a vector of exogenous covariates that accounts for province capacities, locations, and time. A sizable literature (e.g., Newey and Powell, 2003) studies the nonparametric identification and estimation of models in this form. The key requirement is that, in addition to requiring instruments that are mean- or quantile-independent of the unobservables, the instruments should induce sufficient variation in q .

We use demand shifters in (11) as instruments for supply (Wright, 1928). The excluded instruments include wind generation, one-month lagged CDD, and exponentiated demand residuals, which we assume to be orthogonal to shocks to coal production costs (MacKay and Miller, 2025; Döpper et al., 2025). We do not use the shifters as IVs directly. Instead, we construct IVs that more flexibly capture their variation by replacing each continuous variable with a set of indicators for whether its value falls within predefined bins.³⁴ We use these indicators and their interactions with the policy indicator as IVs.

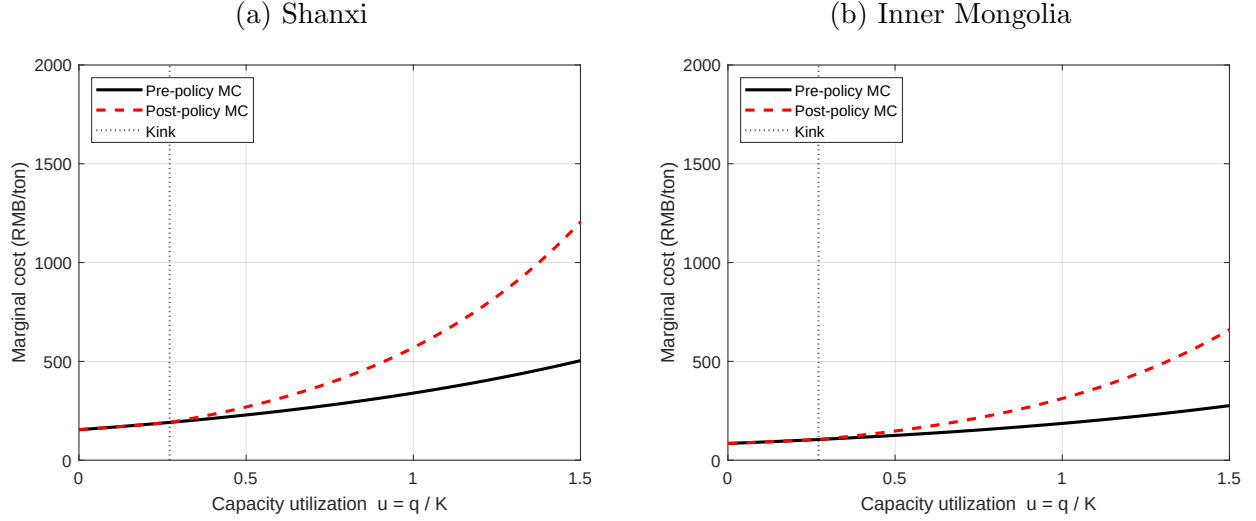
Table 5: Supply Estimation

	(1) Baseline	(2) Lower Threshold	(3) Higher Threshold
α^S	1.014*** (0.216)	1.118*** (0.218)	0.705*** (0.189)
$\alpha^{S,\text{policy}}$	1.007*** (0.252)	0.682*** (0.152)	1.669*** (0.520)
ν	0.523** (0.233)	0.290	0.756
Province FE	YES	YES	YES
Month FE	YES	YES	YES
N	1200	1200	1200

Notes: The table reports GMM estimates of the supply model using indicator-function instruments. Column (1) estimates ν . Columns (2) and (3) re-estimate the slope parameters fixing ν at the column (1) estimate minus and plus one bootstrap standard error, respectively. Standard errors, reported in parentheses, are computed using a province-cluster wild bootstrap. *, **, and *** indicate significance at the 10, 5, and 1 percent levels.

³⁴For wind, we use bins based on 0.25, 0.5, and 1 TWh, which correspond to the variable's quantiles of 32.5%, 52.7% and 79.8%. For CDD, we use 0 and 50 degree days, which correspond to the variable's quantiles of 40.6% and 89.2%. For exponentiated demand residuals, we use 2.5, 5, and 10 thousand tons, which correspond to the variable's quantiles of 10.4%, 51.9%, and 80.2%.

Figure 3: Estimated Aggregate Marginal Costs of the Two Largest Coal-Producing Provinces



Notes: The figures compare the aggregate supply functions of Shanxi and Inner Mongolia with and without the policy effect in 2017.

6.2 Estimation

Column (1) of Table 5 reports the baseline estimates of (16). The policy roughly doubles the slope on quantity above the threshold ν . The threshold is estimated slightly less precisely, so we assess the robustness of other parameters' estimates by fixing ν one standard error below and above the baseline estimate and re-estimating the supply in columns (2) and (3). We also report counterfactual results using these estimates in Appendix K.

To visualize these costs, Figure 3 plots the aggregate marginal cost functions for Shanxi and Inner Mongolia with and without the policy effect, based on the baseline estimates. We interpret these increases in marginal costs as reflecting several factors. First, the policy may impose direct compliance costs, such as halting production for inspections. Second, by preventing low-cost mines from operating above capacity, the policy reallocates output toward higher-cost mines, shifting the production mix toward costlier producers and raising aggregate marginal costs for any given total quantity.

We also compare the average variable costs implied by the baseline estimates with costs reported in the financial statements of publicly listed coal-mining firms in 2014–2016. We plot these costs in Appendix Figure J.5. The estimated and reported costs are well aligned, with a correlation coefficient of 0.563. For example, we find that Xinjiang and Inner Mongolia have the lowest production costs, consistent with more open-pit mining in these provinces.

Finally, although we use a parsimonious specification in which each province's policy effect

scales with its pre-policy aggregate marginal cost, Section 7.2 and Appendix J.5 show that the model’s equilibrium simulations closely match observed prices and output at both the national and provincial levels.

7 Counterfactual Simulation

We begin by defining the equilibrium given a set of shocks. Let $\mathcal{J}$ denote the set of coal-consuming provinces, $\mathcal{I}$ the set of coal-producing provinces, and τ_{ij} the transportation costs.

Definition. In period t , an equilibrium consists of a set of delivery prices $\{p_{jt}\}_{j \in \mathcal{J}}$ and production quantities $\{Q_{it}\}_{i \in \mathcal{I}}$ that satisfy (11) and (17), where province j ’s demand is given by $q_{jt} = \sum_{i \in \mathcal{I}} s_{ij} Q_{it}$, and the price facing producers in province i is $P_{it} = \sum_j s_{ij} (p_{jt} - \tau_{ij})$.³⁵

In all simulations, we allow foreign imports to respond endogenously, following an estimated import supply function.³⁶

The results in this section are based on the baseline demand estimate in column (5) of Table 4 and supply estimate in column (1) of Table 5. In Appendix K, we report the results based on the alternative specifications of supply in columns (2) and (3) of Table 5, which are similar to the baseline results here.

7.1 Economic Surplus and Pollution Damages in the Welfare Function

The reform has two main effects: (1) it reduces capacity by closing small mines and halting new entry, and (2) it restricts production by closely monitoring remaining mines to prevent production above approved capacity. Because we lack mine-level data on entry and exit, we take capacity as given and focus on the effects of production restrictions. Our estimates of economic surplus should be interpreted as the sum of consumer surplus and variable profits, which do not account for investment or fixed costs.

We also compute the pollution damages of coal. We estimate how changes in coal consumption affect air quality, measured by PM2.5 and PM10 concentrations, in each province.

³⁵We can establish equilibrium existence by first constructing closed intervals of prices and quantities on which the equilibrium conditions map the extrema of the intervals back into the intervals, using the monotonicity of the demand and supply functions and the fact that their ranges are $(0, \infty)$. An application of Brouwer’s fixed point theorem then shows existence. For uniqueness, we do not find multiple equilibria in our simulations.

³⁶Our shipment data show that the share of imports to each province is also stable over time, which allows us to construct a similar effective price for imports. We estimate the import supply elasticity in a log-log specification. An OLS regression based on data from 2014 to 2017 (48 months) yields an estimate of 1.35, with a robust standard error of 0.15. A corresponding IV estimation produces nearly identical results.

We then translate the air-quality changes into changes in local health costs and sum them across provinces. In the simulations, these damages vary with both simulated coal consumption and air-pollution shocks, so expected welfare incorporates uncertainty in the pollution damages as well as in demand and supply. We also include environmental costs of coal production based on water depletion. The details are in Appendix I.

Our approach may understate the benefits from reducing coal production because it does not quantify all environmental damages associated with coal. These omitted damages include other forms of pollution from coal production (Chu et al., 2023), transportation, and storage (Jha and Muller, 2018), as well as air pollution from excavation and land degradation.

Despite these issues, we think that our estimates capture the first-order effects of pollution damages. First, reducing coal consumption and the resulting air pollution likely generate the largest welfare gains, given the large affected population, and were plausibly a central objective of the reform (Feng et al., 2019). Second, our estimate of pollution damages during coal production, RMB 86.6 per ton of coal, is larger than the damage value implied by the resource tax. The resource tax is designed primarily to address localized pollution and ecological damage from coal production, and its rate is capped at 10% of coal sales revenue, implying a damage value of RMB 53.2 per ton given the cap and the average coal price in our data, which is below our estimate. Given these considerations, we use the following as the baseline measure of the expected social welfare:

$$W = E(\text{consumer surplus} + \text{producer surplus} + \text{tax revenue} - \text{pollution damages}). \quad (18)$$

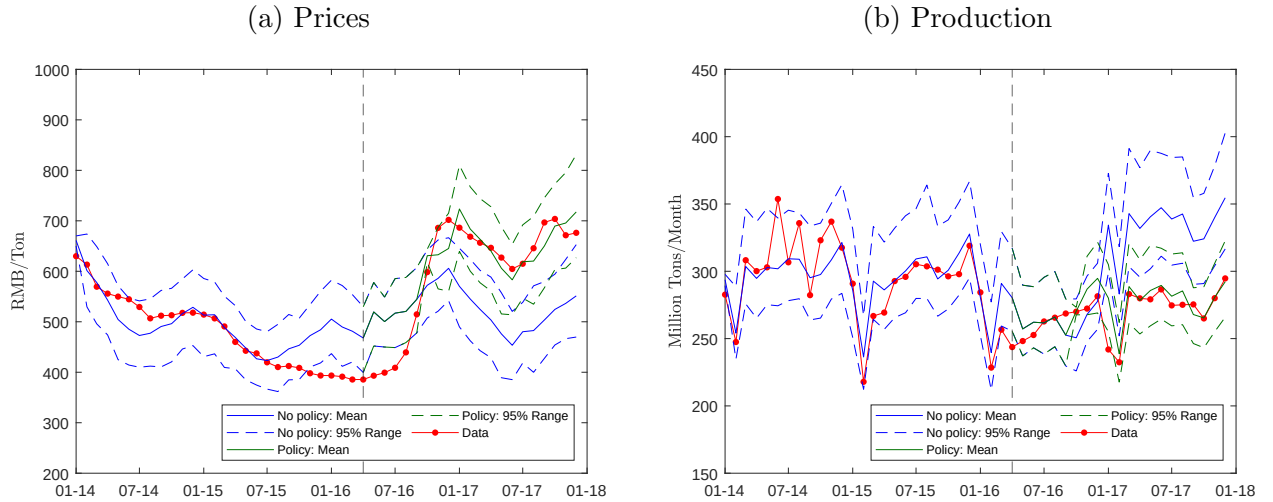
We do not assign an explicit monetary value to carbon emissions in the baseline welfare function. This choice is motivated by both measurement and institutional considerations. Estimates of the social cost of carbon are highly sensitive to assumptions about discount rates, climate damages, and tipping points (Pindyck, 2013), making any single value difficult to justify in this setting. More importantly, climate change is not one of the objectives motivating the 2016 production restriction. We therefore interpret the welfare function in this section as a policy-relevant objective for China’s coal-market intervention, rather than as a complete global social welfare function. This does not mean that damages of carbon are unimportant. Rather, we avoid imposing a particular social cost of carbon in the baseline analysis and instead address these unmeasured damages in Section 7.3.2, where we modify (18) to include a general increasing damage function $L(Q)$ that can capture carbon emissions and other unmeasured social costs of coal.

7.2 Production Restriction

To simulate the no-policy outcomes, we remove the estimated policy effect in supply by setting $\alpha^{S,\text{policy}} = 0$ and recompute the expected equilibrium outcomes. We assume that demand and supply shocks at the province level follow an AR(1) process, and we also simulate forward the processes of CDD and wind generation.³⁷ We report the 2.5%–97.5% range and the average of market outcomes based on 300 simulation paths.

Figure 4 compares equilibrium prices and production across the no-policy simulation, the policy simulation, and the data, holding capacity fixed at observed levels. The simulations closely match the data before and after the policy. Absent the policy, prices would have been substantially lower and production substantially higher. This is also generally the case in major coal-producing provinces (Appendix J.5).

Figure 4: Prices and Production With and Without the Policy



Notes: The figures plot the average national prices and total production from 300 simulations across the no-policy simulation, policy simulation, and the data. We plot the simulated 2.5%–97.5% range and the average from simulations, and the data paths. The vertical lines indicate the start of the policy.

Welfare Effects

Table 6 presents the changes of market outcomes in a no-policy scenario, relative to the observed policy, from the equilibrium simulations. We focus on 2017 because inventory adjustment was unusually large in 2016: between April and December 2016, a substantial inventory drawdown supplemented coal consumption by end users. By 2017, inventory levels

³⁷Appendix G reports the estimates of these processes.

had largely stabilized, so consumption more closely tracked production. This makes 2017 the appropriate period to apply our equilibrium framework, in which demand reflects consumption and supply maps production to consumption without a sizable inventory-change component. We simulate the market outcomes with and without the policy effects using the estimated time-series processes for shocks and other covariates, conditional on their values in the last month of 2016.³⁸

Table 6 shows that removing the policy lowers the delivery price by RMB 145/ton, a 22.1% decrease from the policy level of RMB 656.6/ton, and raises domestic production by 668 million tons, a 20% increase from 3.33 billion tons, in 2017. Consumers gain RMB 560 billion from lower coal prices, while producers lose RMB 288 billion and resource-tax revenue falls by RMB 14 billion. The no-policy scenario raises economic surplus by RMB 259 billion, but pollution damages increase by RMB 309 billion because of higher coal production and consumption. Social welfare is therefore RMB 50 billion lower without the policy. Appendix Figure J.7 breaks down the surplus changes at the province level, consistent with the policy benefiting coal-producing provinces and hurting coal-consuming provinces.

Table 6: No-policy Effects

Δ Delivery price (RMB/ton)	-145 [-157, -132]
Δ Domestic coal production (million tons/year)	668 [589, 762]
Δ Coal consumption (million tons/year)	621 [545, 712]
Δ Economic Surplus and Pollution Damages (RMB bn.)	
Economic Surplus	259 [222, 296]
Consumer	560 [489, 629]
Producer	-288 [-320, -255]
Resource Tax	-14 [-16, -12]
Avoided Pollution Damages	-309 [-341, -282]
Δ Welfare	-50 [-73, -29]

Notes: The table reports changes relative to the observed policy in 2017. Square brackets report the 2.5–97.5 percent range across simulations. Pollution damages include local air-pollution health costs and water depletion.

³⁸It is possible that the time-series processes before and after the policy differ. We find that all of our results are qualitatively robust to using processes estimated based on 2017 data alone.

Allocative Inefficiency

We next quantify the policy’s inefficiency relative to an efficiently implemented quota. We note two types of inefficiencies. First, our estimates suggest that the policy raises infra-marginal costs and leads to within-province allocative inefficiency. Second, our estimates indicate greater output decreases in larger, lower-cost provinces such as Inner Mongolia, which leads to cross-province inefficiency. Although a cap-and-trade system or a productivity-based production restriction is likely infeasible given the institutional constraints, this exercise sheds light on the welfare costs of the policy and the possible policy preferences that may justify it, which we discuss in Section 7.4.

We first decompose the inefficiencies caused by the supply function shift in each province. Holding the 2017 equilibrium quantity fixed in each simulation path, we recompute producer surplus using the pre-policy supply curve integrated up to that quantity. The within-province inefficiency is therefore the increase in producer surplus under the pre-policy supply curve relative to the observed policy, holding fixed the same provincial quantities. This inefficiency is estimated to be RMB 136 billion.

Next, we measure the total inefficiency. An efficient quota allocation across provinces requires the marginal surplus loss from an additional unit of reduction to be equalized across provinces. Therefore, holding fixed a time series of simulated shocks in 2017, for any total reduction ΔQ , the efficient allocation is the one implied by a uniform tax that achieves the same total reduction. We use the following procedure to find the efficient allocation: in each simulation based on the post-policy supply function, we (i) solve for the tax that delivers the same aggregate reduction in equilibrium, (ii) fix the quantities produced in each province under this tax, and (iii) compute economic surplus by integrating demand and the pre-policy supply functions up to those efficient quantities. The average total producer-surplus gain from correcting within- and cross-province misallocation is RMB 212 billion. Thus, within-province misallocation accounts for 64% of total misallocation.

7.3 Is Tax More Efficient?

In Section 2.3.2, we discuss the political reasons why production restrictions, as opposed to a tax, are used to implement the reform. Yet the economic question remains: is it more efficient to use a tax? Section 3 shows that whether the best tax achieves higher welfare than the production restriction depends on the slopes of the demand, cost, and marginal damages, their variances, and the allocative inefficiency of the production restriction. We present two sets of results that point to the better efficiency of tax.

7.3.1 Optimal Tax Maximizing Economic Surplus and Avoided Pollution Damages

An optimal counterfactual tax is set at the beginning of 2017 and maximizes the welfare W in (18), where the expected value of each component is based on the estimated AR(1) processes of shocks and their realized values in the last month of 2016.

Table 7: Welfare Effects (RMB bn.) of Taxes Relative to the Observed Policy

	(1) No Policy	(2) Optimal	(3) Optimal, Weighted
Δ Consumer Surplus	560	-602	-368
Δ Producer Surplus	-288	-660	-615
Δ Tax Revenue	-14	1270	1071
Δ Avoided Pollution Damages	-309	253	164
ΔW	-50	261	251

Notes: The table reports changes in 2017 relative to the observed policy. Avoided pollution damages include local air-pollution health benefits and water depletion. No tax is imposed in column (1). The tax levels in columns (2) and (3) are 417 and 331 RMB/ton, respectively.

Column (2) of Table 7 reports the welfare effects of the optimal tax. The optimal tax is RMB 417/ton and raises W by RMB 261 billion relative to the observed policy. Under this tax, the 2017 delivery price (inclusive of the tax producers pay) increases to RMB 855/ton, consumer surplus falls by RMB 602 billion and producer surplus falls by RMB 660 billion, but these losses are more than offset by over RMB 1 trillion in tax revenue³⁹ and RMB 253 billion in additional avoided pollution damages. Rebating the tax revenue would offset the decline in economic surplus. This result arises because the observed equilibrium is on relatively inelastic portions of both demand and supply. Relative to the no-policy equilibrium in column (1), the economic surplus under the optimal tax is RMB 250 billion lower, but the overall surplus is higher because of greater avoided pollution damages.

Anticipation of shocks. An assumption in the calculation above is that the unobservables in the demand and supply functions are unknown to market participants and policymakers. This assumption is important for correctly defining the expected welfare, but our qualitative results on the efficiency of the tax are likely robust to stronger information assumptions. In the extreme case where the unobservables are fully known to coal buyers, sellers, and the government, a tax would be unambiguously more efficient than a production restriction for the same quantity target.

³⁹We include both the existing resource tax and the counterfactual tax.

Ad valorem tax. Our baseline tax analysis uses a unit tax. The results are robust to instead imposing an ad valorem tax. The optimal counterfactual ad valorem tax rate is 57%, in addition to the existing resource tax. This raises the average 2017 delivery price to RMB 835/ton. The changes in consumer and producer surplus relative to the observed policy are similar to those in column (2) of Table 7: consumer surplus falls by RMB 618 billion and producer surplus falls by RMB 662 billion. The total welfare gain is nearly identical, at RMB 261 billion.

7.3.2 Optimal Tax with a General Damage Function

Next, we significantly strengthen the efficiency comparison using the stochastic-dominance-based criterion from Section 3.5. We construct a more general welfare measure that may account for broader effects of coal consumption, such as carbon emissions, by further including an increasing damage function $L(Q)$ on total output. There may also be other pollution externalities that we cannot model directly but that can be approximated as an increasing function of aggregate production Q .⁴⁰ The baseline welfare measure augmented with this general damage function is

$$\begin{aligned} W^* &= E(\text{consumer surplus} + \text{producer surplus} + \text{tax revenue} - \text{pollution damages} - L(Q)) \\ &= W - E(L(Q)). \end{aligned}$$

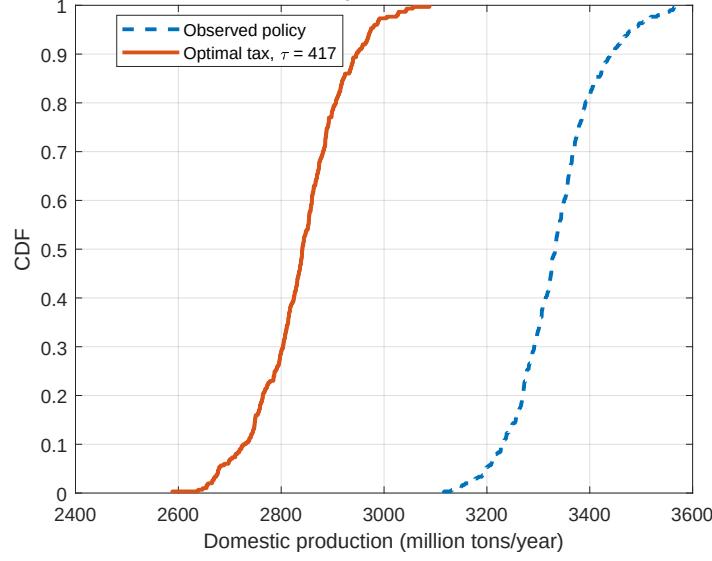
Without specifying L other than assuming that it increases in Q , we apply the criterion in Section 3.5 to compare W^* under the observed policy with that under the optimal tax from the previous section: given that the tax achieves a higher W than the observed policy (column (2) of Table 7), we just need to check whether the quantity distribution under the tax is stochastically dominated. Figure 5 confirms this: the output CDF under the optimal tax lies everywhere above the CDF under the production restriction.⁴¹ Since $W_{\text{tax}} > W_{\text{restriction}}$ and

⁴⁰Leakage is an externality that we do not explicitly account for. After coal (65%), hydro (20%) had the second-largest share in the power generation mix in 2016, followed by wind (4%) and nuclear (4%). Natural gas accounted for 3% of the generation mix (Göß and Niggemeier, 2017). In the long run, reduced coal consumption, especially in electricity generation, is likely to be offset by both the growth of renewable generation and increased natural gas use (EIA, 2022). Generation from these sources typically has lower local pollution intensity and carbon emissions than coal. Therefore, such substitution would still be expected to reduce overall pollution damage when coal consumption falls.

⁴¹There are several ways to formally test the stochastic dominance. First, for a simple but formal statistical test, we test an even stronger condition, namely that quantity under the optimal tax, based on the estimated parameters, is statistically smaller than under the policy for each simulated path of shocks. More generally, given sufficiently many simulation draws, we can ignore simulation error and compute the maximal difference between the two CDFs over their support. The 95% confidence interval of this difference can be simulated using the asymptotic distribution of the estimated parameters.

$E(L(Q_{\text{tax}})) \leq E(L(Q_{\text{restriction}}))$ for any increasing L , the tax achieves a higher W^* . Moreover, a range of taxes around the optimal tax also achieve higher W , with quantity distributions stochastically dominated by that under the production restriction.

Figure 5: Quantity CDFs: Production Restriction and Optimal Tax



Notes: This figure plots the CDFs of 2017 production under the observed policy and the optimal tax (RMB 417/ton).

7.4 Policy Preferences

The results above show that the government was willing to forgo significant welfare gains when choosing the production restriction over the optimal tax. As Sections 2 and 3 show, redistributing tax revenue is not frictionless, which may cause the government to value the increase in producer surplus, a policy priority, more than the increase in tax revenue. Reducing pollution damages is another central policy objective. Therefore, policy preferences for these two components of the social welfare may explain the government's policy choice. To explore this possibility, we modify the baseline welfare function (18) and focus on the following government objective function as a weighted welfare function:

$$\begin{aligned} \tilde{W}(w_{\text{PS}}, w_{\text{damage}}) = & E(\text{consumer surplus} + w_{\text{PS}} \cdot \text{producer surplus} \\ & + \text{tax revenue} - w_{\text{damage}} \cdot \text{pollution damages}). \end{aligned}$$

Section 3 shows that the effects of these weights on instrument choices are theoretically ambiguous. We thus consider a range of weight pairs, $(w_{\text{PS}}, w_{\text{damage}})$. For each pair, we solve

for the tax maximizing $\tilde{W}$ and then evaluate whether $\tilde{W}$ is higher under that tax than under the observed production restriction policy.

We find that, for a given value of w_{damage} , the production restriction generates a higher $\tilde{W}$ than the optimal tax when w_{PS} exceeds a threshold. Table 8 reports these thresholds. When the government values consumer surplus, tax revenue, and pollution damages equally ($w_{\text{damage}} = 1$), valuing producer surplus 40.8% more justifies the observed production restriction relative to even the optimal tax.⁴² Given this policy preference, the tax maximizing government objective function $\tilde{W}$ is RMB 331/ton. This tax achieves the same value of $\tilde{W}$ as the observed policy, but the total unweighted welfare would be RMB 251 billion higher (column (3) of Table 7).⁴³ At any greater value of w_{PS} , $\tilde{W}$ is higher under the observed policy than under even the best tax. This result is consistent with the intuition in Section 3.4 that a stronger preference for producer surplus favors production restrictions, because a tax can cause a weighted producer-surplus loss larger than the value of tax revenue, while restrictions can directly raise producer surplus.⁴⁴

When effective redistribution of tax revenues is infeasible (Section 2), several mechanisms could rationalize a policy preference for producer surplus. First, the profits of coal mining firms, many of which are state-owned, are not entirely private rents. Part of the state-owned firms' after-tax profits flows back to the government, giving the government a direct fiscal stake in coal-firm profitability. Second, maintaining coal producers' profitability helps to stabilize employment and avoid worker resettlement. The falling prices in 2014–2015 forced provincial governments to reduce local taxes and expand the safety net (Shanxi, 2015, 2016).⁴⁵ Third, these profits reduce the need for external financing when the government encourages investment in new upgrades (Yuan, 2025).

⁴²This result is different from some policy preference estimates in other settings. For example, Lim and Yurukoglu (2018) estimates that the government places much higher weight on consumer surplus than on utilities in the utility rate-setting context in the US.

⁴³We find the same minimal value of w_{PS} for the government to prefer the observed policy to any ad valorem tax, and the ad valorem tax maximizing the weighted objective function is 50%, in addition to the existing resource tax.

⁴⁴We can also rationalize the observed policy in terms of the cost of redistributing tax revenue. Let w denote administrative costs and other frictions as a share of tax revenue, so that only $(1 - w)$ of tax revenue enters the government's policy objective. For the government to prefer the observed policy to a tax with the weighted welfare function $E(\text{CS} + \text{PS} + (1 - w)\text{tax} - \text{pollution})$, we find that the minimal w is 27%.

⁴⁵The government also set up a RMB 100 billion fund to compensate the coal and steel workers that were laid off because of the reform (Section 2), but the size of the fund is an order of magnitude smaller than the tax revenue to raise and redistribute under the optimal tax (column (2) of Table 7).

Table 8: Policy Weights That Rationalize Production Restrictions Over Taxes

w_{damage}	1.00	1.25	1.50	1.75	2.00
$w_{\text{PS}} \geq$	1.408	1.491	1.601	1.732	1.878

Notes: The table reports the minimal weight on producer surplus required for the government to prefer the production restriction policy to the optimal tax for a given value of w_{damage} . The optimal tax levels corresponding to the columns are 325, 425, 525, 625, 650 RMB/ton.

We also note that placing a greater weight on pollution damages raises the threshold value of w_{PS} needed to rationalize the observed policy relative to a tax. The intuition is related to our finding in Section 7.3.2: given our demand and supply estimates, a tax can often achieve higher economic surplus while reducing quantities, and therefore pollution damages, more than the production restriction policy. As a result, a tax becomes more attractive at higher values of w_{damage} , so a larger weight on producer surplus is required for the government to prefer production restrictions over a tax.

This result also has an important implication for our interpretation of the lower bound on w_{PS} . As discussed above, our estimates may potentially understate total pollution damages. If the omitted damages are proportional to the damages we measure and enter the government objective function in the same way, then our welfare measures should be adjusted with a greater w_{damage} . This would require a higher w_{PS} to rationalize the production restriction policy. At the same time, the qualitative conclusion that a higher policy preference for producer surplus helps to rationalize the observed policy remains unchanged.

8 Conclusion

We study China’s coal industry reform since 2016, which restricted coal production through mine closures and output controls. Using an estimated equilibrium model of provincial coal demand and supply, we find that the policy reduces coal output, raises prices, and shifts surplus from consumers to producers. We also develop a new approach to compare the observed production restriction with a tax.

Our welfare analysis shows that the production restriction is not the surplus-maximizing way to reduce coal consumption under equal welfare weights in this setting. A tax could have reduced aggregate output while raising total welfare, which accounts for both the economic surplus from coal production and the environmental externalities. We discuss the political constraints that may have limited the feasibility of such a tax.

We translate these constraints into policy preferences. The reform documents emphasize not only pollution reduction, but also coal-industry profitability and economic stability in producing regions. When tax revenue cannot be effectively redistributed, we model the policy objective as a weighted welfare function. We show that even an optimal tax no longer dominates the observed policy when the government places a sufficiently high weight on producer surplus. More broadly, we provide an empirical framework to bound these weights.

The results thus underscore the challenges of transitioning to a low-carbon economy given the institutional constraints. Consistent with this observation, China has made massive investments in renewable energy. In 2024, the share of coal in electricity generation fell below 60% for the first time, a decline driven primarily by the growth of low-cost electricity production from renewable sources, as opposed to reduced coal consumption (Howe, 2024).

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A Policy Documents

Figure A.1 provides a snapshot of the original government document for the reform. We provide an English translation of this document in this section.

In order to implement the “*Opinions of the State Council on Resolving Overcapacity in the Coal Industry and Achieving Sustainable Development*” (Guo Fa [2016] No. 7, hereinafter referred to as the “Opinions”), and to further regulate and improve the coal production and management order, effectively resolve overcapacity, and promote the sustainable development of coal enterprises, the following notice is issued regarding related issues:

1. Fully understand the importance of standardizing and improving the order of coal production and operation

Currently, the coal industry is in a serious predicament. Especially since last year, some coal enterprises have resorted to measures such as overcapacity production, compensating for price through volume, and low-price dumping to maintain production operations in an effort to seize market share. These actions have not only severely disrupted the normal order of coal production and operation but have also exacerbated the imbalance between supply and demand in the market, intensifying the difficulties faced by the industry. If these issues are not promptly curbed, the national goals and tasks related to the development of the coal industry will be difficult to achieve. All regions and enterprises must adopt a strategic perspective that promotes supply-side structural reform and effectively addresses capacity reduction tasks. It is essential to enhance overall awareness and fully recognize the importance of standardizing and improving the order of coal production and operation for facilitating the recovery and development of the coal industry. This requires unifying thought and action with the spirit of the “Opinions”, taking effective measures to ensure the implementation of regulations concerning the control of overcapacity production, reduction of production volume, suspension of production during holidays, and maintenance of fair competition.

2. Main Measures to Standardize and Improve Production and Operation Order

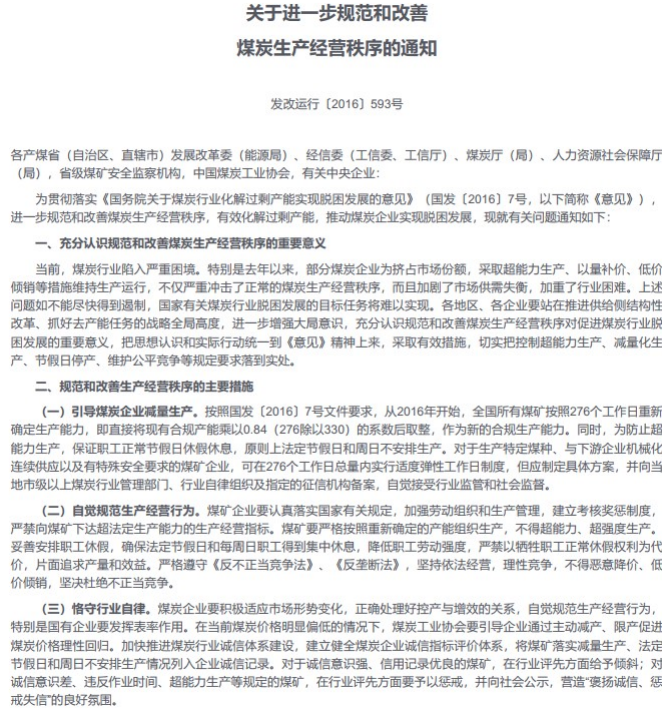
- (a) **Guiding Coal Enterprises to Reduce Production Volume:** In accordance with the requirements of the State Council Document [2016] No. 7, starting from 2016, all coal mines nationwide are to re-determine their production capacity based on 276 working days. This means directly multiplying the existing compliant capacity by a coefficient of 0.84 (276 divided by 330) and rounding to the nearest whole number to establish the new compliant production capacity. Additionally, to prevent

overcapacity production and ensure employees can take their normal holidays, no production should be scheduled on statutory holidays and Sundays. For coal mines producing specific coal types, providing mechanized continuous supply to downstream enterprises, or with special safety requirements, a moderately flexible working day system may be implemented within the total of 276 working days, provided that a specific plan is developed and filed with local coal industry management departments at or above the municipal level, industry self-regulatory organizations, and designated credit institutions, and that these enterprises consciously accept industry regulation and social supervision.

- (b) **Consciously Standardizing Production and Operation Behavior:** Coal mines must earnestly implement national regulations, strengthen labor organization and production management, and establish a performance assessment and reward-punishment system. It is strictly prohibited to set production and operation targets that exceed the legal production capacity for coal mines. Production must be organized strictly according to the re-determined capacity, and overcapacity or excessively intense production is not allowed. Employee leave must be properly arranged to ensure that staff can rest during statutory holidays and every Sunday, thereby reducing labor intensity. It is forbidden to sacrifice employees' normal leave rights in order to pursue output and efficiency. Coal mines must strictly adhere to the "Anti-Unfair Competition Law" and the "Anti-Monopoly Law," operate in accordance with the law, engage in rational competition, refrain from malicious price cuts, and eliminate unfair competition.
- (c) **Upholding Industry Self-Regulation:** Coal enterprises must actively adapt to changes in market conditions, correctly balance production control and efficiency enhancement, and consciously regulate their production and operation behavior, especially state-owned enterprises which should set a leading example. In the current context of notably low coal prices, the coal industry association should guide enterprises to promote rational price recovery through proactive production reduction and output limitations. The construction of an integrity system in the coal industry should be accelerated, establishing and improving an integrity index evaluation system for coal enterprises. Compliance with production reduction, as well as not scheduling production on statutory holidays and Sundays, should be included in the integrity records of enterprises. Coal mines with strong integrity awareness and good credit records should receive favorable treatment in industry evaluations, while those with poor integrity awareness, violations of operational hours, and overcapacity production should face penalties in industry evaluations

and be publicly disclosed to create a positive atmosphere of “rewarding integrity and punishing dishonesty.”

Figure A.1: Policy Document



B Data Comparison

To validate the provincial and monthly production and consumption data we compiled from various sources, we compare the statistics based on our data and those from the National Bureau of Statistics of China in Table B.1. The first column displays the national values aggregated from our data based on coal transportation, the second column is derived from the NBS Coal Balance Table for all provinces, and the third column represents the national statistics in the NBS National Coal Balance Table.

Overall, the national statistics derived from our transportation data are reasonably close to those in NBS. For instance, both datasets show declining output and consumption between 2014 and 2016, and a decrease in imports in 2015 followed by a rebound in 2016.

We also validate our data at the province-year level. Figure B.1 compares the shipment out of and into a province in each year with the annual coal production and consumption data by province in the provincial coal balance tables from NBS. Overall, they align closely,

Table B.1: Data Comparison of Coal Output and Consumption (million tons)

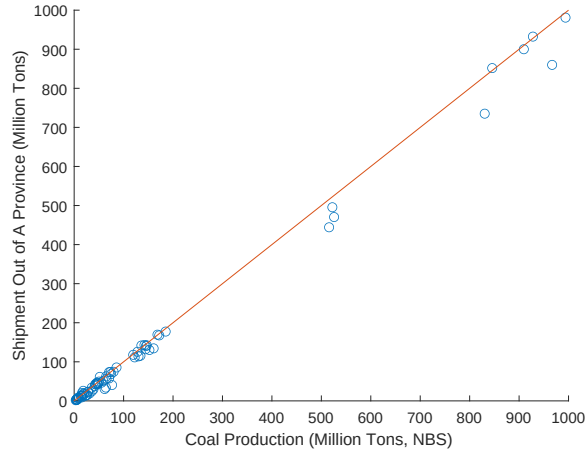
	2014			2015			2016		
	Main	NBS (Prov)	NBS (Nat)	Main	NBS (Prov)	NBS (Nat)	Main	NBS (Prov)	NBS (Nat)
Output	3703	3876	3874	3457	3747	3747	3135	3411	3411
Consumption	3860	4317	4136	3662	4255	3998	3574	4249	3888
Thermal power	1974	1823	1895	1872	1792	1796	1908	1818	1827
Heating supply	181	306	224	204	323	241	222	345	266
Coking Coal	563	667	639	530	625	609	536	626	606

Notes: The table compares the transportation data used in the empirical analysis (under main columns) with the annual numbers reported by the National Bureau of Statistics of China. The column of NBS (Prov) contains data collected from province coal balance tables and aggregated to national numbers, and the column NBS (Nat) uses statistics reported in the national coal balance tables. The sum of coal consumption across provinces exceeds the national total because of the potential double counting according to the official Q&A from China NBS (See https://www.stats.gov.cn/hd/cjwtd/202302/t20230207_1902276.html). Under the main column, the output is calculated as total shipment, and the monthly consumption is constructed in Appendix C. We then aggregate province-month production and consumption to the national level.

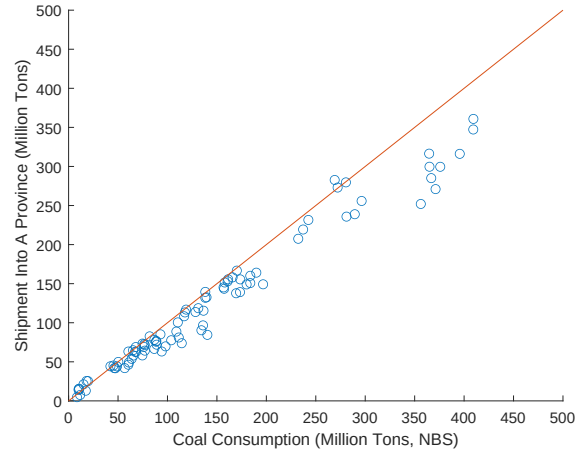
with the imputed consumption data being slightly lower than those reported by the NBS. This difference is expected because NBS data explicitly state that they do not remove certain instances of double-counting at the province level.⁴⁶

Figure B.1: Data Comparison of Shipment, Coal Output and Consumption at Province Levels (Million Tons)

(a) Shipment Out of a Province vs. Production



(b) Shipment into a Province vs. Consumption



Notes: The figure compares province-year coal production and consumption in the data used in our empirical analysis with those reported by province coal balance tables. One dot in the figures represents one province-year pair. The y -axis is calculated by aggregating monthly total shipment out of and into a province within a year.

C Coal Consumption

We collect monthly coal consumption data from 2014 to 2016 across seven sectors: electricity, coking, construction, heating supply, steel-making, chemistry, and others. Table C.1 displays annual consumption in these sectors. Electricity is the largest coal consumer, accounting for over 50% of total consumption. Other major users include coking and cement plants, each accounting for 15%. The consumption breakdown is stable during our sample period. The “others” sector, which mainly includes residential demand and small-scale industrial processes, accounted for 7% in 2014 and declined to 3% in 2016. We plot the national time series of the downstream output in Figure C.1.

⁴⁶For example, coal transported to province A for washing and burned at power plants in B are both counted as consumption in provinces A and B. Our shipment data only count the final consumption in B.

Table C.1: Consumption Shares by Sectors

Year	2014	2015	2016
Electricity	0.51	0.51	0.53
Coking	0.15	0.14	0.15
Construction	0.15	0.15	0.14
Heating Supply	0.05	0.06	0.06
Steel-making (steam coal only)	0.04	0.04	0.04
Chemistry	0.04	0.04	0.04
Others	0.07	0.06	0.03

Notes: The table shows annual coal consumption shares by sector. We collected monthly coal consumption by sector from the firm China Coal Resource, and then aggregated them annually to calculate those shares.

To allocate coal consumption by sector across provinces, we further collect the monthly output of primary products in each sector by province from the NBS and convert it into coal consumption. For the electricity sector, we use thermal electricity output as the main product. Although thermal electricity generation includes both coal and natural gas, natural gas contributes to only about 4.5% of total thermal generation, making its impact negligible.⁴⁷ We use cement as the representative product for the construction sector, as it accounts for 83.7% of coal consumption there. In the chemistry sector, fertilizers serve as the primary product.

Provincial coal balance tables provide annual coal consumption for thermal electricity generation and coking. We aggregate thermal and coke output to the province-year level and calculate coal consumption per 1 MWh of electricity and per ton of coke for each province and year. We assume that these ratios remain constant across months within a province and year, converting thermal electricity generation and coke into coal consumption for the electricity and coking sectors. For cement, steel, and fertilizers, we aggregate their output by month across provinces and calculate the coal-to-output ratio. Assuming this ratio is constant across provinces within a month, we compute coal consumption in the construction, steel, and chemistry sectors based on their province-month output. Table C.2 shows that producing 1 MWh of electricity uses less coal over time, reflecting the gradual improvement in power plant energy efficiency. The conversion ratios are also consistent with engineering estimates by industry experts.⁴⁸

⁴⁷<https://chinaenergyportal.org/en/2016-detailed-electricity-statistics-updated/>

⁴⁸For example, producing 1 ton of cement requires 0.11 ton of standard coal, equivalent to a conversion ratio of 0.15 between coal and cement, given that 1 ton of raw coal equals 0.7143 ton of standard coal. See <https://www.chinanews.com.cn/cj/2011/05-03/3011403.shtml>.

Table C.2: Coal-to-End Product Output Ratio by Sectors

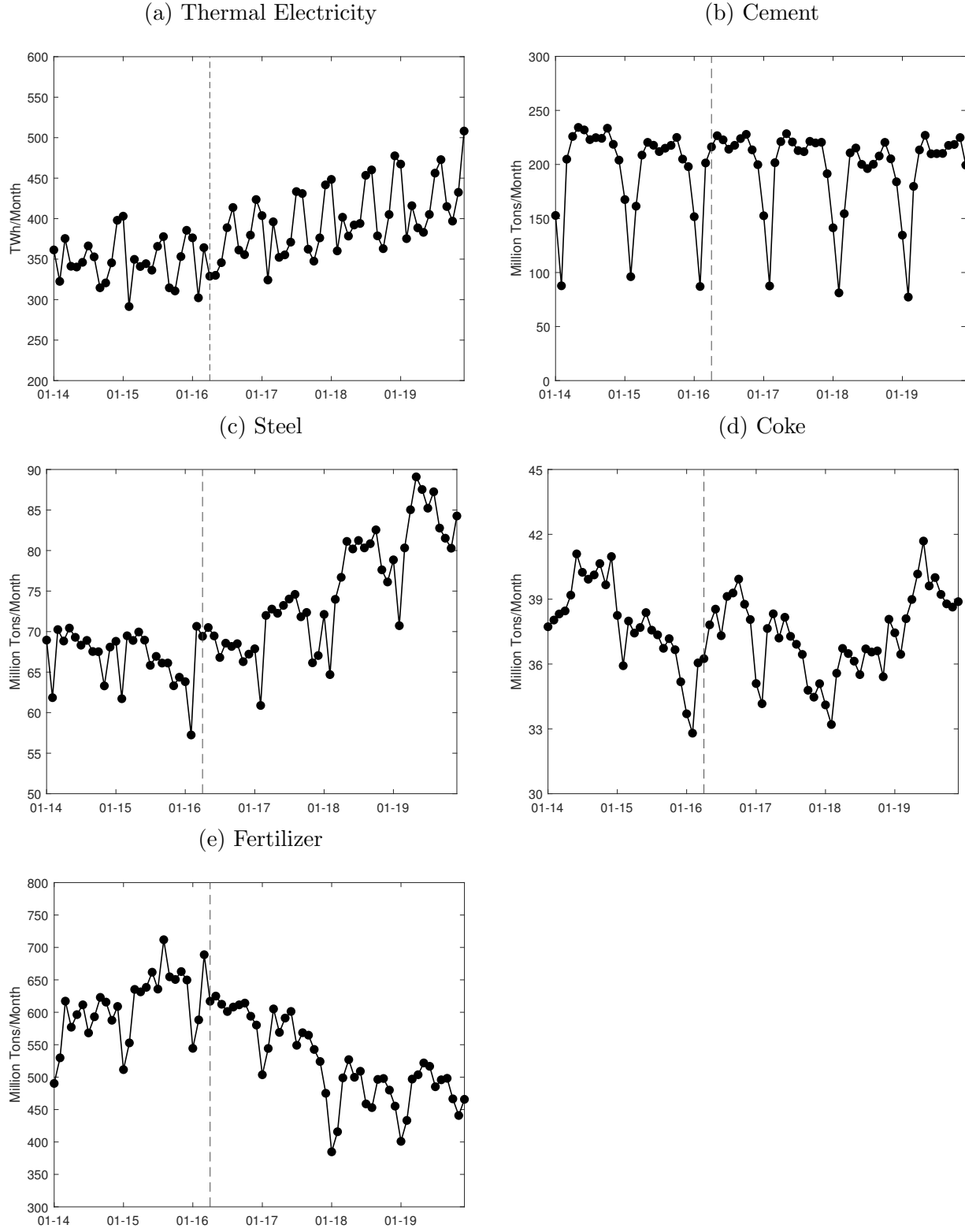
Year	2014	2015	2016
Electricity (Ton of coal/MWh)	0.47	0.44	0.43
Cement (Ton of coal/Ton)	0.23	0.24	0.21
Steel (Ton of coal/Ton)	0.14	0.12	0.13
Coke (Ton of coal/Ton)	1.18	1.19	1.20
Fertilizer (Ton of coal/Ton)	2.01	1.85	2.00

Notes: The table reports the conversion ratios used to estimate coal consumption for the production of key downstream products in each sector. Monthly data on the output of thermal electricity, cement, steel, coke, and fertilizers by province is collected from the NBS. The provincial coal balance tables provide annual coal consumption for thermal electricity generation and coking. We aggregate thermal and coke output to the province-year level to calculate the coal consumption per 1 MWh of electricity and per ton of coke. The table shows the average across provinces for each year. For cement, steel, and fertilizers, we calculate national output annually and match it with coal consumption data for the construction, steel (steam coal only), and chemical sectors from China Coal Resource, resulting in an annual conversion rate applied uniformly across all provinces.

For the heating sector and the “other” sector, which lack representative products, we collect annual coal consumption for heating supply from provincial coal balance tables. We then compute annual consumption in the “other” sector as the residual of total coal consumption after accounting for electricity generation, heating supply, industry, and construction. Using these data, we first calculate province-specific annual coal consumption shares by sector and then allocate monthly consumption across sectors according to these shares.

We extrapolate these data to 2017. Our monthly, sector-level consumption data and cross-province shipment data end in 2016. For 2017, we impute the consumption data using the province-level monthly production data from NBS. Specifically, we allocate each province’s production to destination provinces using shipment shares observed in 2016. We note that the shipment quantity is slightly lower than NBS’s production data in prior years, likely due to a double-counting issue with the NBS data as discussed in Appendix B. Therefore, the allocated production is scaled down by 0.936, the average production-to-shipment ratio observed in 2014–15. We assume that the total production allocated to a province is its consumption, which is a reasonable assumption given that coal inventory has stabilized in 2017 (Figure 2).

Figure C.1: Output in Major Downstream Sectors



Notes: Panel (a) plots the national thermal electricity generation, (b) plots the national cement production, (c) plots the national steel production, (d) plots the national coke production, (e) plots the national fertilizer production.

D China's Major Infrastructure Development in April and May of 2016

The list below represents our best-effort search for major active infrastructure projects (with a cost over RMB 50 billion) in April and May of 2016, based on NDRC announcements, Xinhua News, China Daily and other news sources. We focus on projects that are extremely material-intensive, thus requiring significant use of coal and coke for concrete and steel. We exclude the Guangzhou–Shenzhen–Hong Kong Express Rail Link (Hong Kong section), which meets the RMB 50 billion cost threshold, but the scale (26km of railway) was far smaller than the other projects.

New Projects

1. Yuxi–Mohan Railway (China-Laos Railway) in Yunnan (southwest of China). Full-scale construction began on April 19, 2016, following earlier preparatory works that started in September 2015. The project's total cost is about RMB 51.6 billion.⁴⁹
2. Beijing–Zhangjiakou High-Speed Railway (HSR) in Beijing and Hebei (central China). The construction started on April 29, 2016 (Wang, 2021), with a total investment of RMB 58.4 billion.⁵⁰
3. Tianfu International Airport in Chengdu (southwest of China). The construction started in May 2016, with a total investment of RMB 72 billion.⁵¹

Ongoing Projects

1. Haoji Railway between Shanxi and Inner Mongolia (northwest-central China). The construction started in 2015 and was completed in 2019, with a cost of more than RMB 193 billion.⁵²
2. Daxing International Airport in Beijing and Hebei (central China). The construction started in 2014 and was completed in 2019, with a cost of RMB 80 billion.⁵³

⁴⁹<https://global.chinadaily.com.cn/a/202007/03/WS5efe84baa310834817256ef1.html>

⁵⁰<https://www.reuters.com/article/sports/china-approves-9-2-billion-winter-olympics-high-speed-rail-line-local-government-idUSKCN0SS04F/>

⁵¹<https://aviationweek.com/air-transport/chinas-new-chengdu-airport-facilitates-massive-air-travel-growth>

⁵²<https://global.chinadaily.com.cn/a/201910/05/WS5d97d87fa310cf3e3556ecde.html>

⁵³https://www.chinadaily.com.cn/business/2014-12/26/content_19177195.htm

3. Beijing–Shenyang HSR in Beijing, Hebei and Liaoning (central-north China). The construction began in 2014 and was completed in 2021, with a cost of RMB 123.5 billion.⁵⁴
4. Hong Kong–Zhuhai–Macau Bridge in Guangdong, Hong Kong and Macau (south China). The construction began in 2009 and was completed in 2018, with a cost of RMB 127 billion.⁵⁵
5. Wudongde Hydropower Station in Sichuan and Yunnan (southwest of China). The construction started in 2015 and was completed in 2021, with a cost of RMB 120 billion.⁵⁶
6. Lianghekou Hydropower Station in Sichuan (southwest of China). The construction started in 2014 and was completed in 2022, with a cost of RMB 66.5 billion.⁵⁷

E Derivations of Equations in Section 3

Derivation of Equation (4)

A production restriction modifies the marginal abatement cost C' in a neighbourhood of $\hat{q}$ by raising it by $L'(\hat{q})$ at $\hat{q}$ and making the private marginal cost steeper. Therefore C'' is raised to $C'' + \kappa$. With the shock α , the equilibrium quantity deviation x satisfies the approximate first-order condition

$$C'(\hat{q}) - L'(\hat{q}) + (C'' + \kappa)x + \alpha = 0, \quad x \equiv q - \hat{q}.$$

At the social optimum, $C'(\hat{q}) = B'(\hat{q}) = L'(\hat{q})$, so the level terms cancel:

$$(C'' + \kappa)x + \alpha = 0.$$

Therefore

$$x_{\text{restriction}} = -\frac{\alpha}{C'' + \kappa}.$$

⁵⁴https://usa.chinadaily.com.cn/china/2013-12/30/content_17205381.htm

⁵⁵https://web.archive.org/web/20190327195326/http://www.xinhuanet.com/english/2018-10/23/c_137553194.htm

⁵⁶<https://www.reuters.com/business/energy/chinas-giant-wudongde-hydro-project-put-into-full-operation-2021-06-16/>

⁵⁷<https://www.chinadailyhk.com/hk/article/264174>

Derivation of Equation (6)

Substituting $x_{\text{restriction}} = -\alpha/(C'' + \kappa)$ into $\Delta W = (\beta - \alpha)x - \frac{D}{2}x^2$, where $D \equiv C'' + \kappa - B''$, and taking expectations (using $\mathbb{E}[\alpha] = \mathbb{E}[\beta] = 0$ and independence):

$$\mathbb{E}[(\beta - \alpha)x_{\text{restriction}}] = \frac{\sigma_\alpha^2}{C'' + \kappa}, \quad \mathbb{E}[x_{\text{restriction}}^2] = \frac{\sigma_\alpha^2}{(C'' + \kappa)^2}.$$

Hence

$$\mathbb{E}[\Delta W_{\text{restriction}}] = \frac{\sigma_\alpha^2}{C'' + \kappa} - \frac{D \sigma_\alpha^2}{2(C'' + \kappa)^2} = \frac{(C'' + B'' + \kappa) \sigma_\alpha^2}{2(C'' + \kappa)^2},$$

using $2(C'' + \kappa) - D = C'' + B'' + \kappa$. Since $\mathbb{E}[\Delta W_{\text{tax}}] = (C'' + B'') \sigma_\alpha^2 / (2(C'')^2)$ from (5),

$$\mathbb{E}[W_{\text{tax}} - W_{\text{restriction}}] = \frac{(C'' + B'') \sigma_\alpha^2}{2(C'')^2} - \frac{(C'' + B'' + \kappa) \sigma_\alpha^2}{2(C'' + \kappa)^2}.$$

Derivation of Equation (10)

Setting $w_{\text{PS}} = 1$ and differentiating with respect to w_{damage} , the envelope theorem gives

$$\frac{\partial \mathbb{E}[\widetilde{W}_{\text{tax}} - \widetilde{W}_{\text{restriction}}]}{\partial w_{\text{damage}}} = \mathbb{E}[L_{\text{restriction}} - L_{\text{tax}}].$$

Expanding L around $\widehat{q}$,

$$\mathbb{E}[L_{\text{restriction}} - L_{\text{tax}}] \simeq L'(\widehat{q}) \mathbb{E}[x_{\text{restriction}} - x_{\text{tax}}] + \frac{L''(\widehat{q})}{2} \mathbb{E}[x_{\text{restriction}}^2 - x_{\text{tax}}^2].$$

Both quantity deviations are mean-zero, so $\mathbb{E}[x_{\text{restriction}} - x_{\text{tax}}] = 0$ and the L' term drops out. Using

$$\mathbb{E}[x_{\text{tax}}^2] = \frac{\sigma_\alpha^2}{(C'')^2}, \quad \mathbb{E}[x_{\text{restriction}}^2] = \frac{\sigma_\alpha^2}{(C'' + \kappa)^2},$$

the result follows:

$$\frac{\partial \mathbb{E}[\widetilde{W}_{\text{tax}} - \widetilde{W}_{\text{restriction}}]}{\partial w_{\text{damage}}} = -\frac{L''}{2} \left(\frac{\sigma_\alpha^2}{(C'')^2} - \frac{\sigma_\alpha^2}{(C'' + \kappa)^2} \right).$$

F Coal Transportation

An integrated railroad and coastal marine shipping system is the primary mode of coal transportation in China, accounting for 65.2% of total coal transport in 2017.⁵⁸ The transportation costs typically include the direct expenses associated with freight charges, which are influenced by the distance to be traveled, the mode of transportation (such as rail, truck, or barge), and the specific logistics involved. While shipping fares for waterway transport are influenced by market forces, railroad transportation fares are typically regulated.⁵⁹ Additionally, loading and unloading fees, terminal charges, and any necessary handling costs contribute to the total transportation expenditure. Other factors, such as fuel prices, maintenance of transportation vehicles, and labor costs, also play a significant role in shaping these costs.

Due to the limited variation in transportation shares, directly estimating transportation costs through shipping choices via a behavioral model is challenging. Instead, we manually collected estimated transportation costs between major coal producers and consumers from an authoritative industry report (Cao and Feng, 2017) published by China Bond Rating Co., Ltd., a leading agency that investigates and collects industry information to assess default risks of bonds across various sectors, including coal. Then we fit a quadratic relationship between the transportation costs and the distance between any two provinces, imposing the restriction that transportation costs increase with distance up to the maximum distance between any two provinces in China. Figure F.1 shows the transportation costs of province pairs estimated by the industry report, along with the estimated relationship between inter-provincial distance and transportation costs. The intercept represents the costs of transporting coal within a province, estimated to be RMB 60.9/ton. The estimates also indicate slight distance discounts.

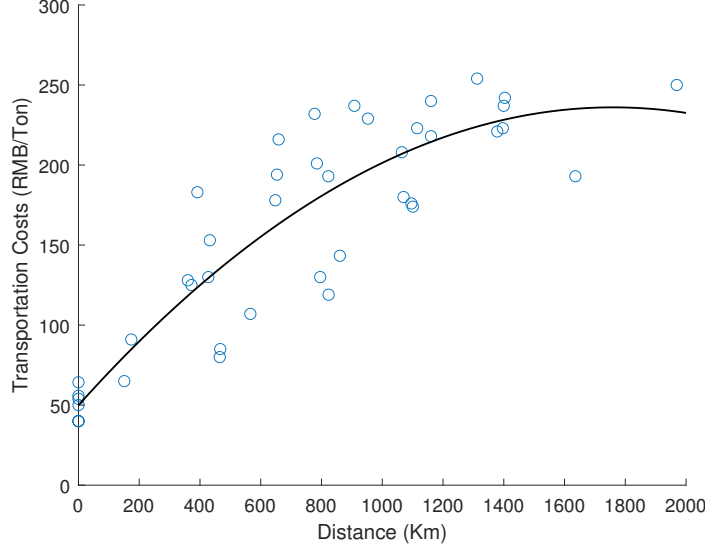
⁵⁸See http://pdf.dfcfw.com/pdf/H3_AP201811151246522585_1.PDF.

⁵⁹For more details about the composition of regulated fares for transporting coal by railroad, please refer to <http://www.gov.cn/xinwen/2017-12/26/5250421/files/5a71188c4fd34d1d9f05ef5e16690d2e.pdf>.

Table G.1: Estimated AR(1) Process for Demand and Cost Shocks

	Demand	Domestic Supply
δ_1	0.805 (0.025)	0.494 (0.066)
N	1020	850

Figure F.1: Estimated Transportation Costs



Notes: The figure shows transportation costs (RMB/ton) between major coal-producing and coal-consuming provinces, plotted against the distance between their capital cities (km), based on an industry report (Cao and Feng, 2017) published by China Bond Rating Co., Ltd. We then approximate the relationship between transportation costs and distance using a quadratic function (black curve).

G Simulation of Shocks

We estimate the following AR(1) processes for demand shocks ε_{jt} and supply shocks ξ_{it} :

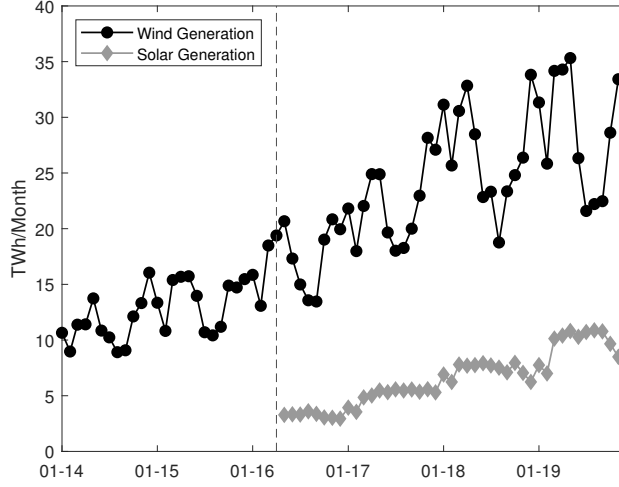
$$\varepsilon_{jt} = \delta_0^D + \delta_1^D \varepsilon_{jt-1} + \omega_{jt}^D$$

$$\xi_{it} = \delta_0^S + \delta_1^S \xi_{it-1} + \omega_{it}^S,$$

Table G.1 reports the estimated AR(1) coefficients, showing that demand shocks exhibit greater persistence than cost shocks.

To forecast future demand, we additionally model the evolution of wind generation and cooling degree days (CDD). Motivated by the growth pattern of the total wind generation at

Figure G.1: Wind and Solar Generation in China



Notes: The figure plots the total monthly wind and solar generation in China. The solar generation data are available after April 2016, and their levels are considerably lower than the wind's.

the national (Figure G.1) and province level, we model the generation as a province-specific linear time trend with province-month-of-the-year fixed effects and shocks:

$$X_{jt}^{\text{Wind}} = \delta_{jm(t)}^{\text{Wind}} + \delta_{j1}^{\text{Wind}} t + \omega_{jt}^{\text{Wind}},$$

which is estimated separately for each province. We assume that $\omega_{jt}^D, \omega_{it}^S, \omega_{jt}^{\text{Wind}}$ are mutually independent in the forward simulation. We do the same for the air pollution shocks in Appendix I. We re-sample historical CDDs by year, under the assumption that cross-year climate variation over the sample period is stable.

H Long-Term Contracts and Demand Elasticities

An important assumption in Section 5 is that the monthly coal price in a province is determined in the market equilibrium. In this section, we explore the robustness to this assumption by examining the sensitivity of demand elasticities when a share of demand is locked in by the contracts.

Specifically, we assume that long-term contracts, often signed at the beginning of a year, commit buyers and sellers to trading a fixed quantity at a fixed price for the coming year. This is the original intent of the long-term contracts, although, as Section 2 shows, few buyers and sellers were compliant between 2014 and 2016. To see how the potential contractual lock-in affects estimated demand elasticities, we consider the following robustness analyses: we

assume that a share of $x_{jn}\%$ of observed quantity demanded in year n and province j , Q_{jn}^* , is the contracted amount. This amount is purchased in equal installments over the 12 months. Therefore, the coal demand in the spot market in Equation (11) becomes

$$\ln(q_{jt}(p_{jt}) - \frac{Q_{jn}^*}{12}x\%) = \alpha^D \ln p_{jt} + \beta_X X_{jt} + F E_j^{\text{demand}} + F E_{m(t)}^{\text{demand}} + \varepsilon_{jt} \quad (\text{H.1})$$

For simplicity, we assume that the share is uniform across provinces $x_{jn} \equiv x_n$ and months, but differs across years. In Table H.1, we report the estimates of the spot market elasticity α^D and the quantity-weighted average adjusted elasticity based on the total quantity for two sequences of x_n . In panel A, we assume that the fixed quantity share is 7.4% for 2014–2016, and is 57.3% for 2017. These estimates are based on the discussion in Section 2. In reality, the locked-in amounts likely differ across months in a year, especially in 2014–2016. Our simple specification still allows us to test whether our demand functional form and estimates are robust to the lock-in effects from such contracts. In panel B, we exclude 2017, the year with the highest contract compliance rate. We find that the implied elasticities in both panels are not statistically distinguishable from the baseline estimates in Section 5.

Table H.1: Locked-Contract Demand: Quantity-Weighted Average Elasticities

<i>Panel A: 2014–2017</i>				
	Policy	SOE	Small-Mine Capacity	Policy+SOE
α^D	-0.747*** (0.206)	-0.673*** (0.204)	-1.353*** (0.505)	-0.689*** (0.197)
Adj. avg. elasticity	-0.613*** (0.169)	-0.552*** (0.167)	-1.110*** (0.414)	-0.565*** (0.161)
<i>Panel B: 2014–2016</i>				
	Policy	SOE	Small-Mine Capacity	Policy+SOE
α^D	-0.427** (0.189)	-0.391** (0.197)	-0.981** (0.398)	-0.404** (0.185)
Adj. avg. elasticity	-0.398** (0.176)	-0.365** (0.184)	-0.914** (0.371)	-0.377** (0.173)

Notes: The table reports locked-contract demand estimates for two samples. Panel A uses annual contract shares [0.074, 0.074, 0.074, 0.573] for 2014–2017. Panel B uses [0.074, 0.074, 0.074] for 2014–2016. Each estimate is from the log spot-demand regression using the indicated IV set. The adjusted average elasticity accounts for fixed quantities by scaling the spot-demand price coefficient by the spot-demand share in total demand for each province-month and then taking the quantity-weighted average across province-month observations. All specifications exclude Beijing and include province and month fixed effects plus wind generation and cooling degree days. The standard errors based on wild bootstrap are clustered at the province level.

I Coal Consumption and Pollution Damages

To quantify the environmental impact of reducing coal production, we additionally model air pollution as a function of coal consumption and health benefits from changes in pollution levels. We also include the damage from coal production based on water depletion.

I.1 Air Pollution

We specify a mapping from coal consumption to monthly PM2.5 and PM10 concentrations within a province,⁶⁰ which are the primary predictors of health costs in the literature we use later. We estimate the following model for the concentration of pollutant $\ell \in \{\text{PM2.5, PM10}\}$ in province j and month t (where the coefficients are ℓ -specific):

$$\ln \ell_{jt} = \psi_C \ln \tilde{C}_{jt} + \psi_X X_{jt} + F E_j^{\text{pollution}} + F E_{m(t)}^{\text{pollution}} + \omega_{jt}. \quad (\text{I.1})$$

In the above, to account for spillover effects (Fu, Viard and Zhang, 2022), we define the variable $\tilde{C}_{jt}$ as the total coal consumption in province j and all of its neighboring provinces in period t .⁶¹ The variable X_{jt} includes meteorological conditions, including temperature, dew point temperature, sea level pressure, wind speed, sky condition total coverage code, precipitation duration and depth. Additionally, we control for fixed effects at the province and month-of-the-year levels.

Coal consumption and the unobservable ω_{jt} may be correlated. Specifically, the coal consumption could be correlated with other economic activities that also contribute to pollution, and also correlated with heightened monitoring and regulation that reduces pollution. We therefore use the same IVs as the demand estimation for coal consumption.⁶²

Table I.1 indicates that a 1% increase in coal consumption in a province and its neighbors raises local PM2.5 concentration by about 1.4%. The effect is similar when we use PM10 concentration as the outcome variable.⁶³

⁶⁰We use the average of all monitoring stations' concentration readings in a province.

⁶¹We do not further normalize the variables by the province's size because both the pollution measure and the consumption are logged, and we control for province fixed effects.

⁶²We construct instruments for $\tilde{C}$ by summing the instruments across neighboring provinces.

⁶³Ito and Zhang (2020) estimate the elasticity of PM10 concentration with respect to coal usage to be 0.5 using a regression discontinuity design exploiting the Huai-River heating policy. This is similar to the OLS estimates from our province-month panel data, but smaller than the IV estimates.

Table I.1: The Effects of Coal Consumption on Air Pollutants

	(1) ln(PM2.5) OLS	(2) ln(PM10) OLS	(3) ln(PM2.5) IV	(4) ln(PM10) IV
$\ln \tilde{C}_{jt}$	0.416*** (0.115)	0.386*** (0.096)	1.438*** (0.332)	1.284*** (0.314)
Province FE	YES	YES	YES	YES
Month FE	YES	YES	YES	YES
Meteorological Controls	YES	YES	YES	YES
Observations	1,440	1,440	1,440	1,440
Adjusted R ²	0.654	0.605	0.554	0.490
Kleibergen-Paap F Statistic			35.13	35.13

Notes: The table reports estimates of the effects of coal consumption on PM2.5 and PM10 concentrations by OLS and IV. All estimates control for meteorological conditions, including air temperature, dew point, sea level pressure, wind speed, sky condition coverage, and liquid precipitation duration and depth. The instruments used are the same as those in estimating demand. Standard errors are clustered at the province level.

I.2 Monetizing Pollution Damages

We monetize morbidity and mortality costs associated with changes in ambient pollutant concentrations following Barwick et al. (2026).⁶⁴ Specifically, they estimate that a 10 $\mu\text{g}/\text{m}^3$ reduction in daily PM2.5 generates \$9.2 billion in annual morbidity benefits. They also quantify mortality costs using a back-of-the-envelope calculation based on the mortality estimates in Ebenstein et al. (2017) (with data from 2004 to 2012). This calculation implies mortality costs of approximately \$13.4 billion per year associated with a 10 $\mu\text{g}/\text{m}^3$ increase in PM10. We convert all estimates to a per-person per-month basis and express them in 2016 RMB.⁶⁵ In the simulation, we first predict the percentage changes of concentrations of both pollutants from coal consumption changes based on the marginal effects of coal consumption estimated using IV, and then translate the level changes to the total morbidity and mortality health costs per capita. The total changes are calculated by weighting the changes by the

⁶⁴An alternative is based on Ito and Zhang (2020), who estimate a household’s willingness-to-pay of 1.34 dollars per year for 1 $\mu\text{g}/\text{m}^3$ of decrease in PM10 based on data from 2006 to 2014. They also derive an income-dependent willingness to pay based on their estimates of a random-coefficient logit model for air purifiers. This measure leads to a slightly lower monetary value for reducing air pollution.

⁶⁵The health cost estimates reported in Barwick et al. (2026) are expressed in 2015 U.S. dollars. We convert these estimates to RMB using an exchange rate of RMB 6.5 per USD. To obtain per-capita values and express them in real 2016 terms, we use a national population of 1.375 billion and consumer price indices (CPI) of 615.2 in 2015 and 637.5 in 2016. Finally, we divide the resulting annual per-capita values by 12 to obtain monthly measures.

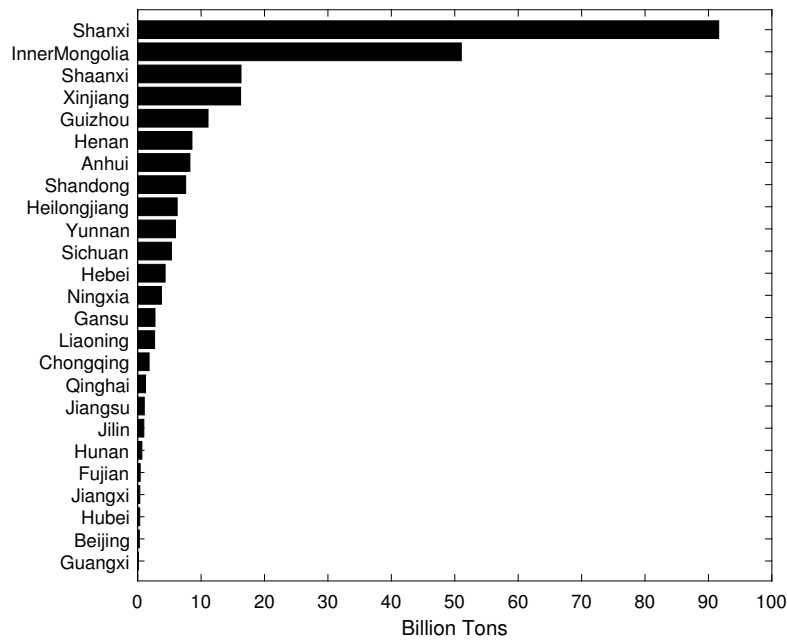
population for each province and then summing across provinces.

We account for the environmental costs of coal production at RMB 86.6 per ton based on the valuation of water depletion (Gu and Li, 2017).

J Additional Figures and Tables

J.1 Distribution of Coal Reserves

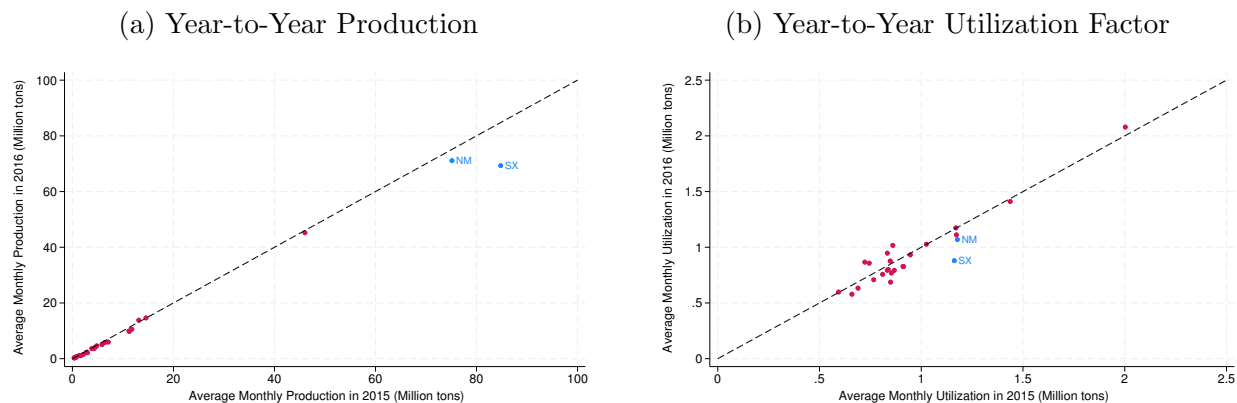
Figure J.1: Coal Reserves in 2016



Notes: Reserves are coal resources that have been surveyed and are feasible for excavation based on data from Qianzhan Industrial Research Institute.

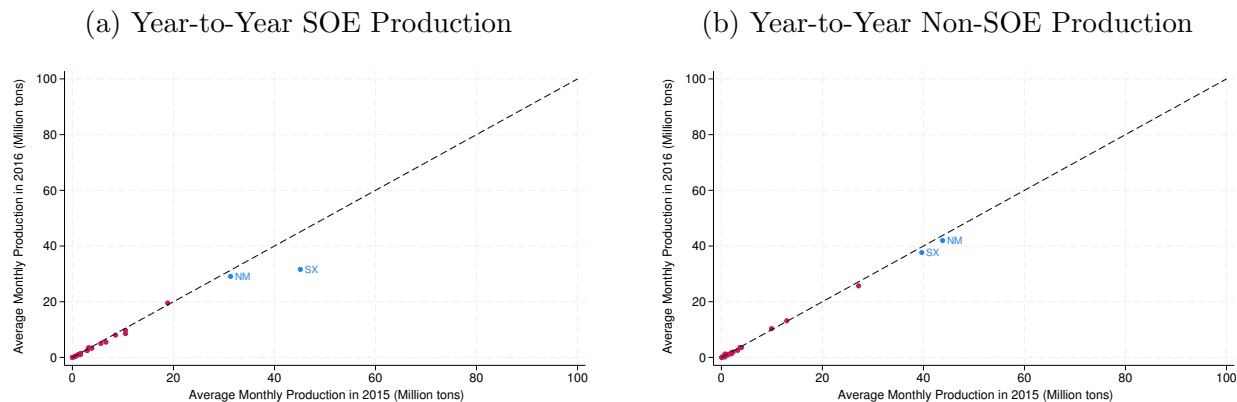
J.2 Policy Effects on Each Province and SOEs

Figure J.2: Shanxi and Inner Mongolia Reduce Production More in 2016



Notes: In panel (a), each dot represents a combination of 2015 (horizontal) and 2016 (vertical) average monthly production of a province. A location below the 45-degree line indicates that the 2016 production is lower. In panel (b), each dot represents a combination of average monthly utilization factors, defined as production over capacity (both normalized to the monthly level), of a province. A location below the 45-degree line indicates that the 2016 utilization factor is lower.

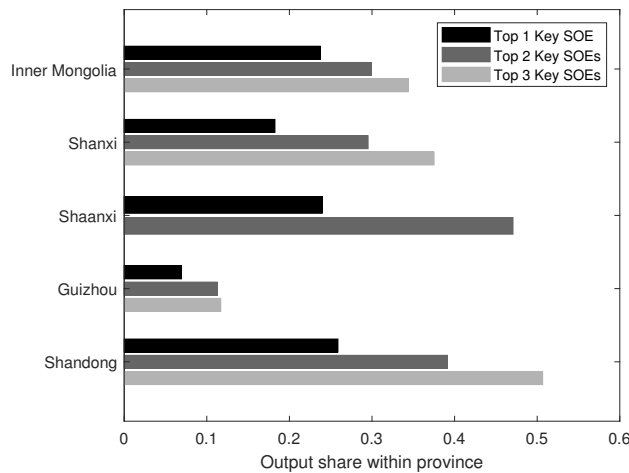
Figure J.3: SOE Reduces Production More in 2016



Notes: In panel (a), each dot represents a combination of 2015 (horizontal) and 2016 (vertical) average monthly production of a province's state-owned enterprises (SOE). A location below the 45-degree line indicates that the 2016 production is lower. In panel (b), each dot represents a combination of average monthly production by a province's non-SOE.

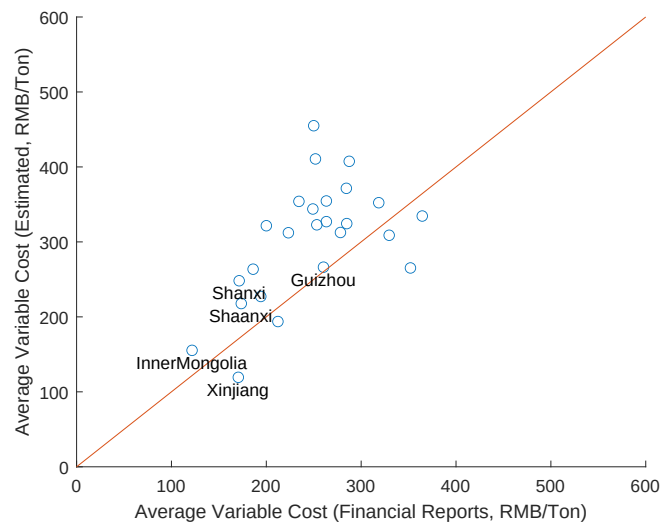
J.3 Concentration of Top State-owned Coal Companies

Figure J.4: Market Shares of Major SOE Coal Producers in Key Provinces



J.4 Comparison between Estimated Average Costs and Unit Costs from Financial Reports

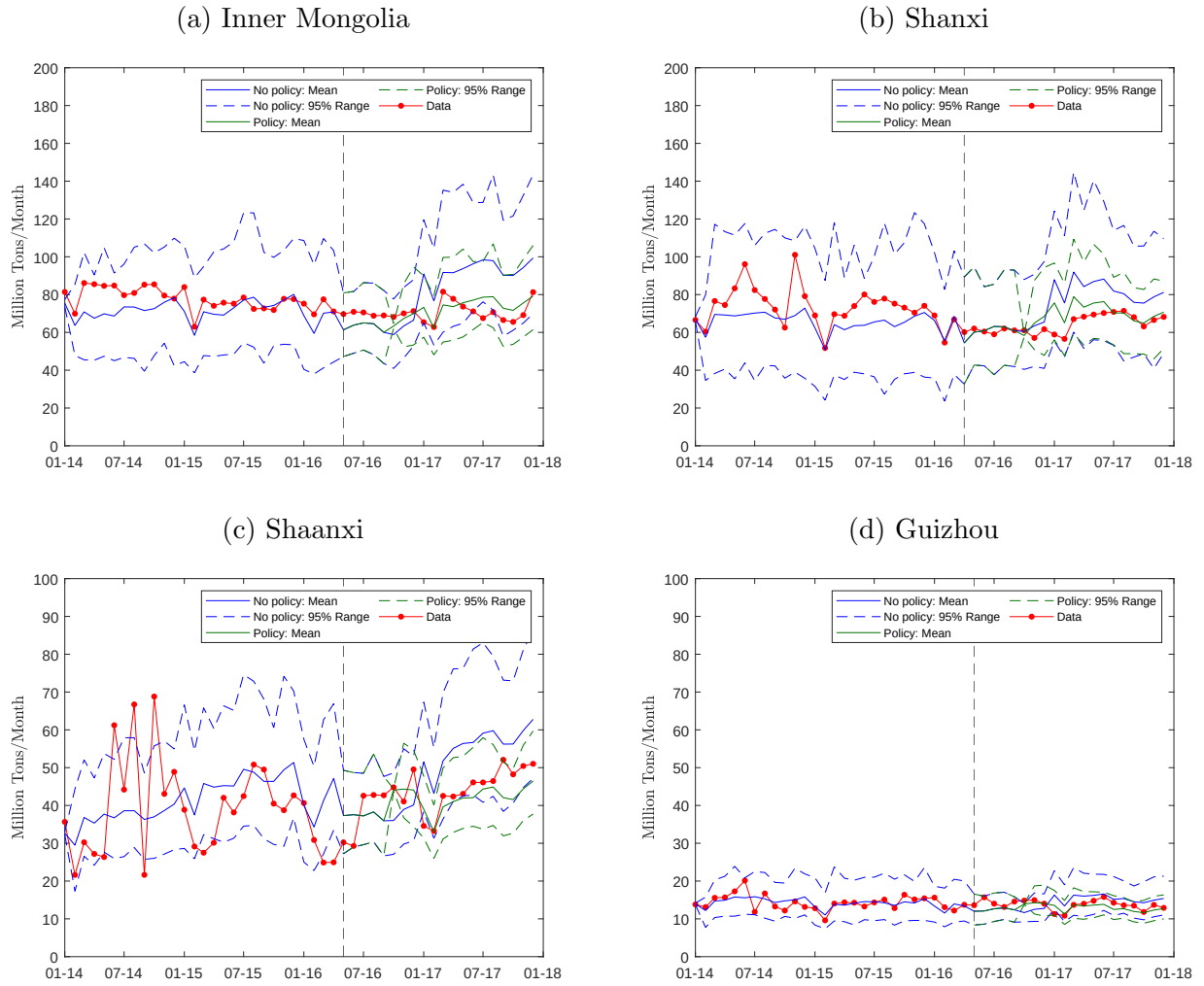
Figure J.5: Compare Estimated Average Costs with Unit Costs from Financial Reports



Notes: The figure compares average costs by province from the estimated model with unit costs from financial reports. The model's estimated average costs are calculated at the observed quantity and averaged between 2014 and 2016. The unit costs are collected from financial reports and bond prospectuses of major coal companies in each province, then averaged across all companies within the province over the same three years in the sample.

J.5 Model Fit by Provinces

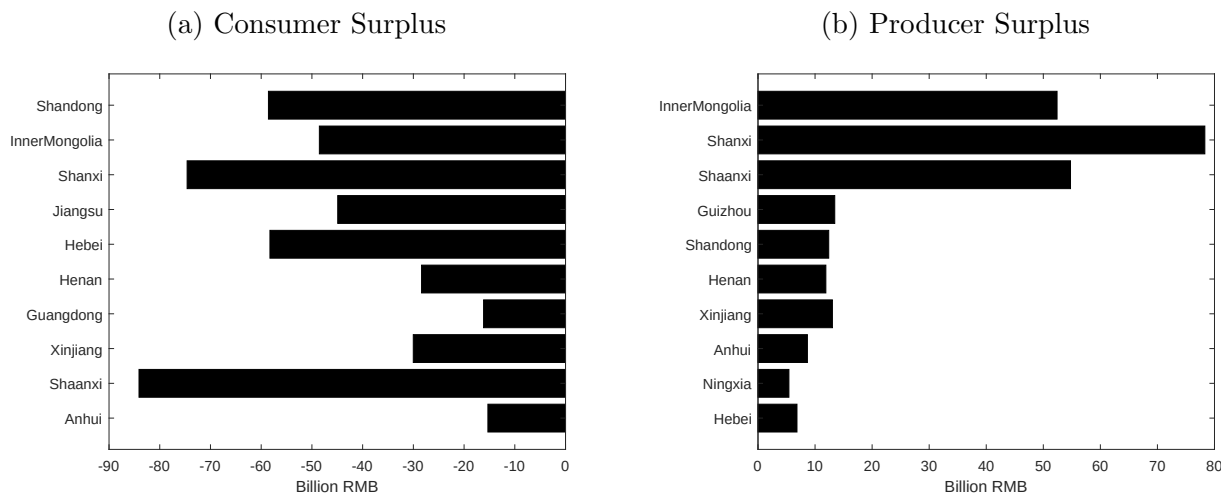
Figure J.6: Simulations: Output by Provinces



Notes: Figures plot the mean and 2.5%–97.5% range across all simulations for output in the four largest coal-producing provinces: Inner Mongolia, Shanxi, Shaanxi and Guizhou.

J.6 Distribution of Economic Surplus by Top 10 Provinces

Figure J.7: Policy Effects on Economic Surpluses in Major Provinces



Notes: The figures illustrate changes in consumer surplus and producer surplus in 2017 relative to a no-policy counterfactual. Changes in consumer surplus are ordered by provinces' average pre-policy consumption, while changes in producer surplus are ordered by provinces' average pre-policy production.

K Threshold Robustness

Table K.1: No-policy Effects: Threshold Robustness

	Lower Threshold	Higher Threshold
Δ Delivery price (RMB/ton)	-137 [-148, -126]	-158 [-177, -139]
Δ Domestic coal production (million tons/year)	626 [569, 697]	753 [628, 906]
Δ Coal consumption (million tons/year)	582 [525, 652]	701 [581, 845]
Δ Welfare (RMB bn.)	-9 [-33, 15]	-101 [-129, -74]

Notes: The table reports changes in 2017 relative to the observed policy. The welfare accounts for changes in economic surplus, tax revenues, and avoided pollution damages. Pollution damages include air-pollution health costs and water depletion. The lower and higher threshold models use estimates from the supply models based on columns (2)–(3) in Table 5.

Table K.2: Policy Inefficiency Relative to Efficient Quotas: Threshold Robustness

	Lower Threshold		Higher Threshold	
	Within-Province	Total	Within-Province	Total
Δ Producer Surplus (RMB bn.)	174	238	91	194

Notes: The table reports producer surplus changes in 2017 relative to the observed policy if we correct the within-province and total misallocation caused by the policy. The lower and higher threshold models use estimates from the supply models based on columns (2)–(3) in Table 5.

Table K.3: Tax Effects (RMB bn.) Relative to the Observed Policy: Threshold Robustness

	Lower Threshold			Higher Threshold		
	(1) No Policy	(2) Optimal	(3) Weighted	(4) No Policy	(5) Optimal	(6) Weighted
Δ Consumer Surplus	530	-612	-270	610	-637	-473
Δ Producer Surplus	-236	-632	-557	-349	-643	-621
Δ Tax Revenue	-13	1278	974	-15	1259	1129
Δ Saved Pollution Damages	-290	257	125	-348	263	203
ΔW	-9	291	271	-101	242	237

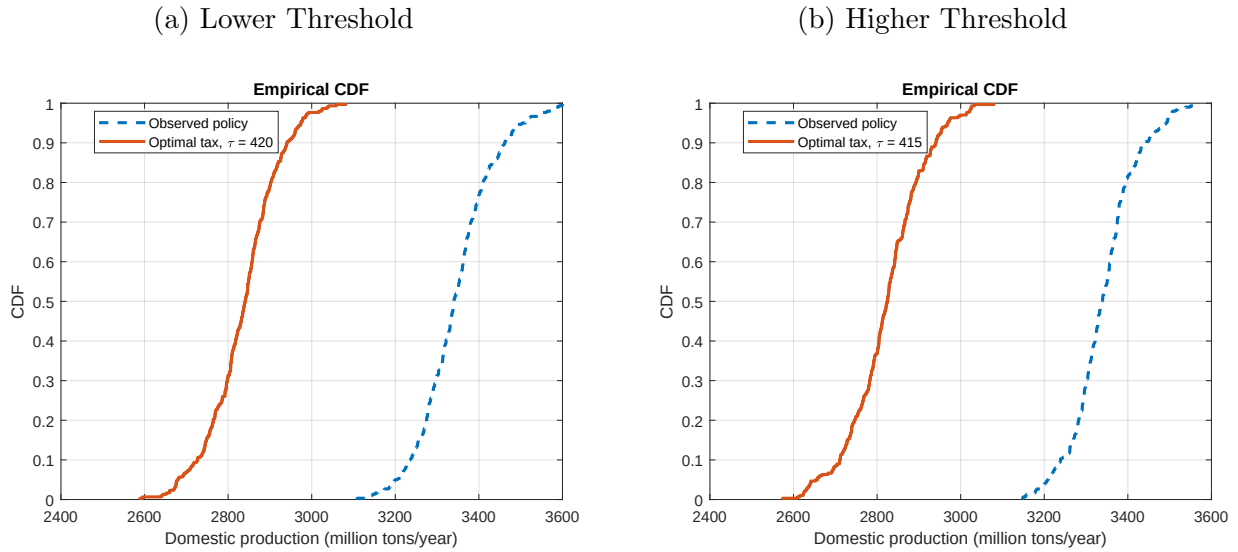
Notes: The table reports changes in 2017 relative to the observed policy. Pollution damages include local air-pollution health costs and water depletion, and exclude carbon. The tax levels in columns (1), (2), and (3) are 0, 420, and 293 RMB/ton under the lower threshold specification, and 0, 415, and 357 RMB/ton under the higher threshold models in columns (4), (5), and (6). The lower and higher threshold models use estimates from the supply models based on columns (2)–(3) in Table 5.

Table K.4: Policy Weights That Rationalize Production Restrictions Over Taxes: Threshold Robustness

w_{damage}	1.00	1.25	1.50	1.75	2.00
Lower Threshold: $w_{\text{PS}} \geq$	1.486	1.563	1.671	1.801	1.950
Higher Threshold: $w_{\text{PS}} \geq$	1.382	1.481	1.607	1.753	1.914

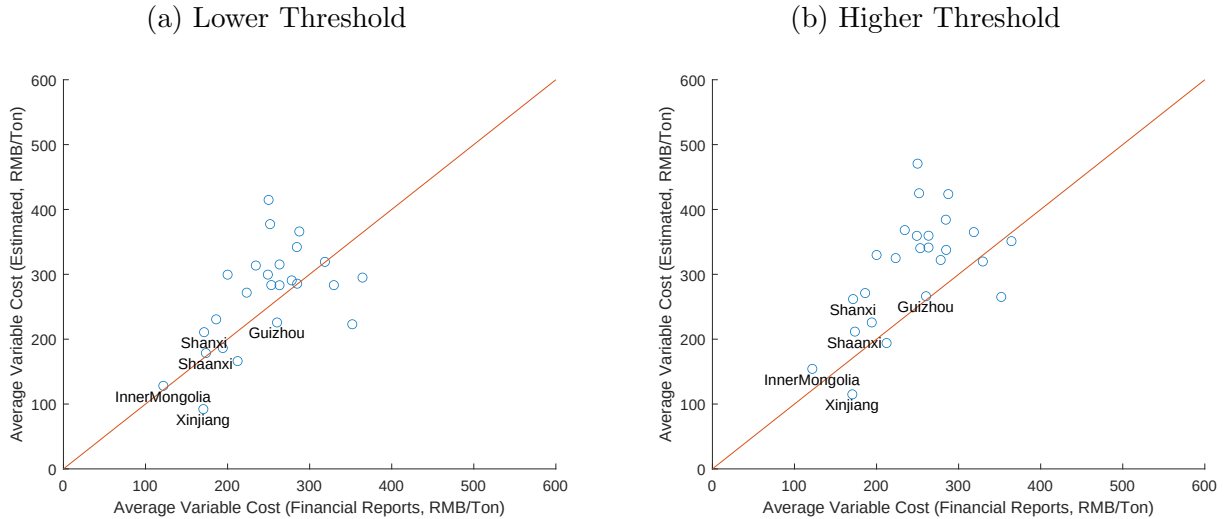
Notes: The best-tax levels corresponding to the columns are: 300, 400, 475, 575, 650 for Lower Threshold; 350, 450, 550, 625, 700 RMB/ton for Higher Threshold. The lower and higher threshold models use estimates from the supply models based on columns (2)–(3) in Table 5.

Figure K.8: Quantity CDFs: Production Restriction and Optimal Tax: Threshold Robustness



Notes: Figures plot the CDFs of 2017 production under the observed policy and the respective optimal taxes using the estimates from the supply models based on columns (2)–(3) in Table 5.

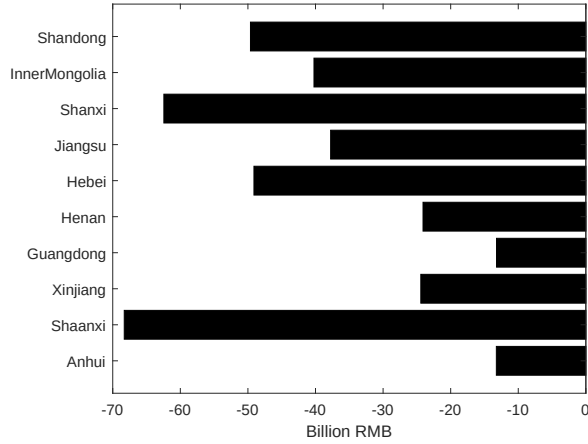
Figure K.9: Compare Estimated Average Costs with Unit Costs from Financial Reports: Threshold Robustness



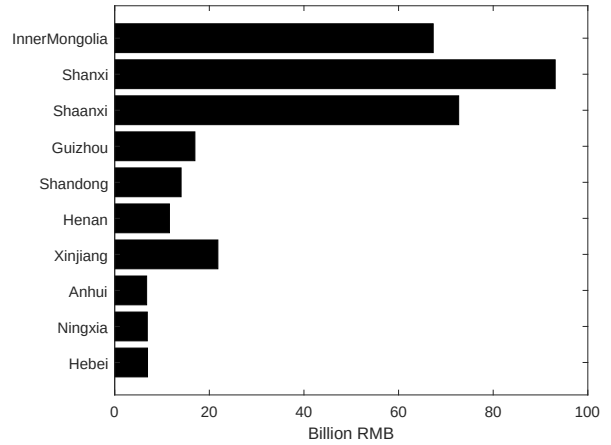
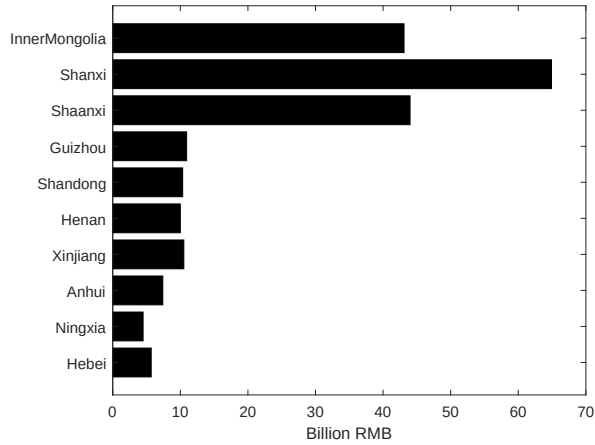
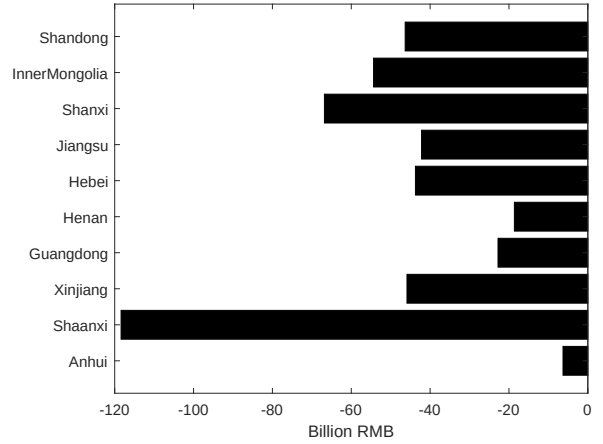
Notes: The figures compare average costs by province from the supply models based on columns (2)–(3) in Table 5 with costs from financial reports. The model's estimated average costs are calculated at the observed quantity and averaged between 2014 and 2016. The unit costs are collected from financial reports and bond prospectuses of major coal companies in each province, then averaged across all companies within the province over the same three years in the sample.

Figure K.10: Welfare Breakdown: Threshold Robustness

(a) Lower Threshold



(b) Higher Threshold



Notes: The figures illustrate changes in consumer surplus and producer surplus in 2017 relative to a no-policy counterfactual when using a lower or higher threshold in the supply model (columns (2)–(3) in Table 5). Changes in consumer surplus are ordered by provinces' average pre-policy consumption, while changes in producer surplus are ordered by provinces' average pre-policy production.